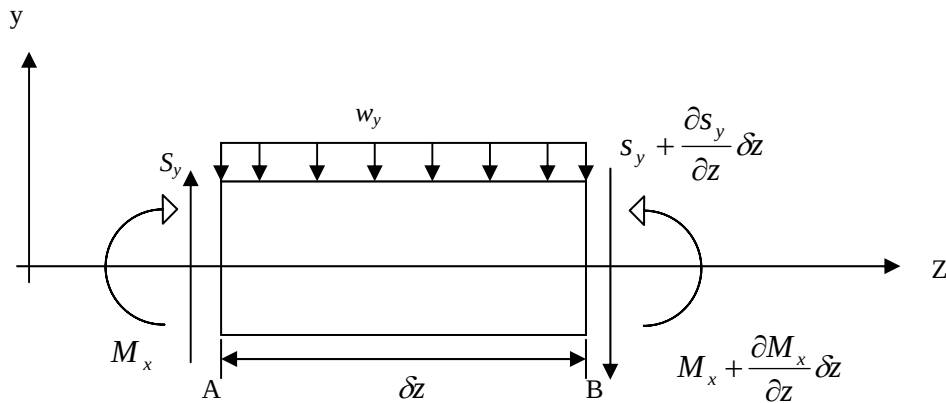


Shear Stress, Bending moment & Displacement Relationships

Load Intensity, Shear Force, Bending Moment, Slope and Deflection of beams are related with the following relationships. Here, the intention is not to derive these relationships but to state them just for information and later use. These relationships can best be demonstrated by considering the equilibrium of the beam segment, shown below.



$$(S_y + \frac{\partial S_y}{\partial z} \delta z) + w_y \delta z - S_y = 0$$

$$\therefore w_y = - \frac{\partial S_y}{\partial z}$$

Taking moment about the mid point of the Left hand edge hand edge , we have:

$$- (M_x + \frac{\partial M_x}{\partial z} \delta z) + (S_y + \frac{\partial S_y}{\partial z} \delta z) \delta z + w_y \frac{(\delta z)^2}{2} + M_x = 0$$

Ignoring second order terms:

$$\therefore S_y = \frac{\partial M_x}{\partial z}$$

If we combine them into a single expression we get:

$$- w_y = \frac{\partial S_y}{\partial z} = \frac{\partial^2 M_x}{\partial z^2}$$

Similarly for load in the XZ plane

$$-w_x = \frac{\partial S_x}{\partial z} = \frac{\partial^2 M_y}{\partial z^2}$$

From simple bending theory we can relate the bending moment to lateral displacement by the following expressions.

$$M_x = EI \frac{\partial^2 y}{\partial z^2}$$

Or

$$M_y = EI \frac{\partial^2 x}{\partial z^2}$$

Integrate to calculate Slope

$$Slope = EI \frac{\partial y}{\partial z}$$

Or

$$Slope = EI \frac{\partial x}{\partial z}$$

Integrate again to calculate the lateral displacements x and y at any point on the beam.

The general relationship is

$$-w_y = \frac{\partial S_y}{\partial z} = \frac{\partial^2 M_x}{\partial z^2} = EI \frac{\partial^4 y}{\partial z^4}$$

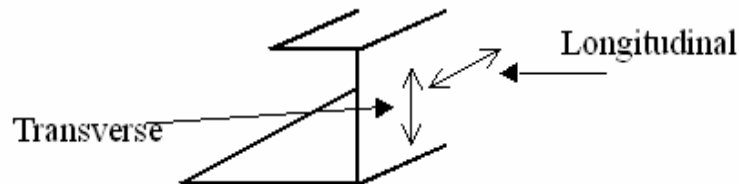
Or

$$-w_x = \frac{\partial S_x}{\partial z} = \frac{\partial^2 M_y}{\partial z^2} = EI \frac{\partial^4 x}{\partial z^4}$$

General shear stress for open and single cell thin walled closed section beams.

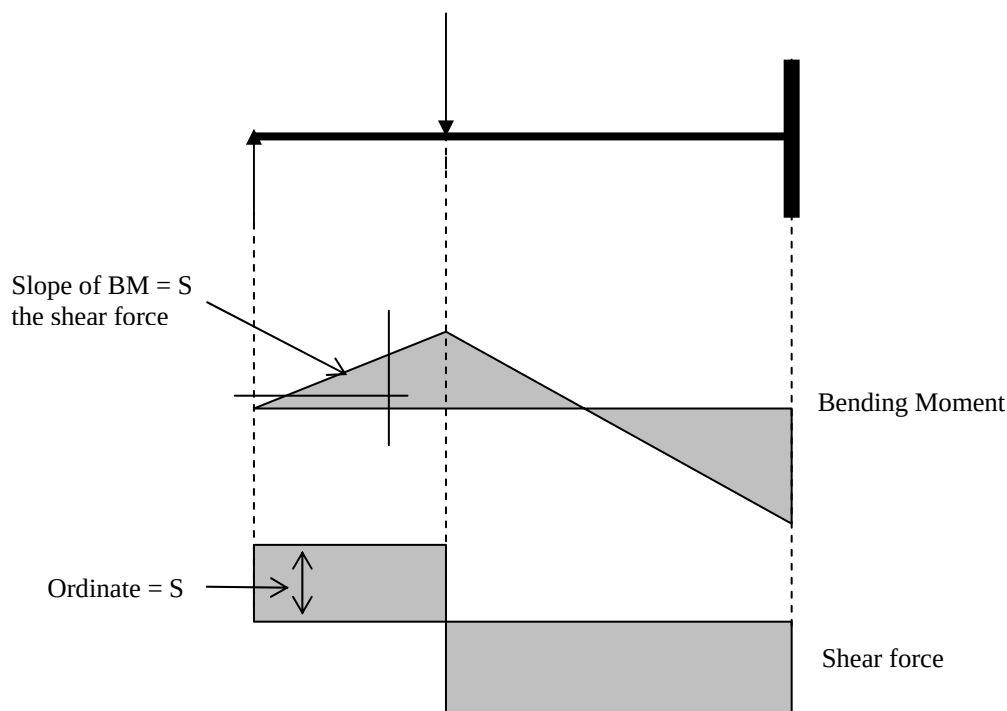
This section deals with analysis of open sections supporting a shear force and closed section beams supporting shear force and/or a torque.

There are many misconceptions about shear. The principal one being that it is primarily a transverse issue - where it is really a longitudinal one!



For a beam we can draw B.M. and S.F. diagrams, which tell us about the local external shear forces but they don't tell us anything about the internal behaviour. For that we need to understand the effects of external shear on the internal shear flow and stress distribution.

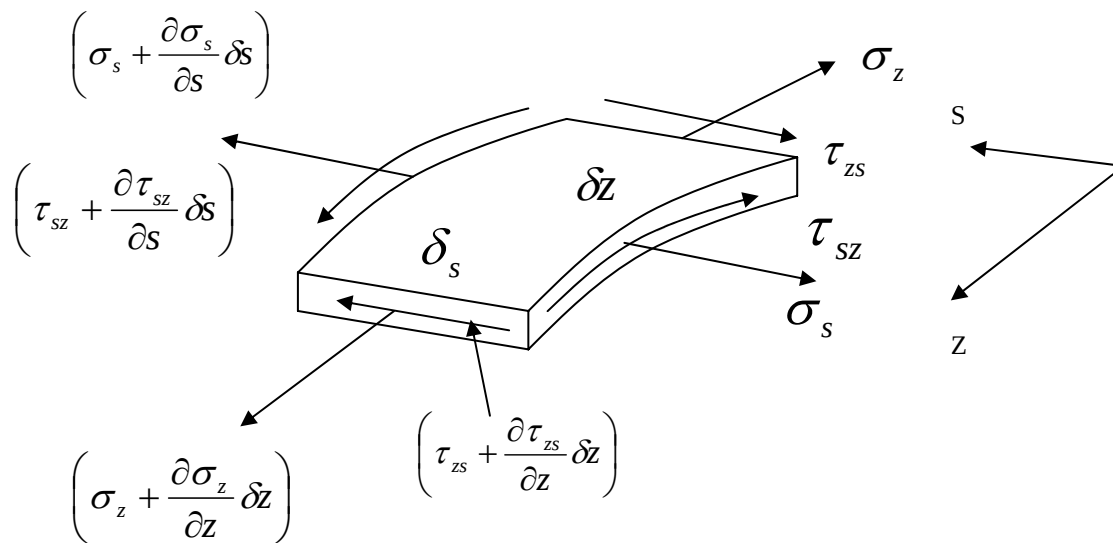
The idea here is to determine the resulting shear flow or shear stress distribution throughout the section a because better understanding of it should result in better and lighter designs.



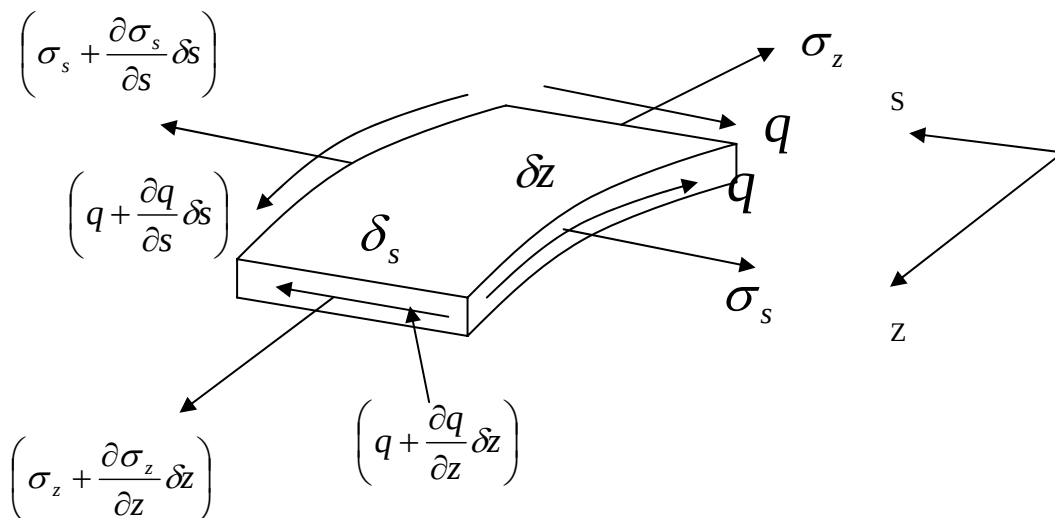
General assumptions:

- Sections are thin, thickness t is small compare to other dimensions
- t^2 and higher order terms are neglected in the calculations
- No shear stress perpendicular to outer surface
- Stresses through the thickness are constant
- Thickness is constant at a section but can vary along the length.

To begin with we must consider the equilibrium of the element of the material shown below:



Or in terms of the shear flow q ,



Consider the equilibrium of the element $\delta s \times \delta z \times t$, shown above, of the beam wall. It may be more convenient to work in terms of q the shear flow, which is the shear force per unit length.

Hence $q = \tau \times t \times 1$ positive in the positive S direction

From the equilibrium of forces in the z direction we have:

$$\left(\sigma_z + \frac{\partial \sigma_z}{\partial z} \delta z \right) t \delta s - \sigma_z t \delta s + \left(q + \frac{\partial q}{\partial s} \delta s \right) \delta z - q \delta z = 0$$

Which reduces to:

$$\boxed{\frac{\partial q}{\partial s} + t \frac{\partial \sigma_z}{\partial z} = 0}$$

Similarly for equilibrium in the s direction:

$$\boxed{\frac{\partial q}{\partial z} + t \frac{\partial \sigma_s}{\partial s} = 0}$$

This is an important equation showing the relationship between direct stress and shear flow or shear stress.

Shear of an open section beams

The open section beam of arbitrary section supports shear loads S_x and S_y through its shear centre. The shear flow and direct stress acting on an element of the beam are related by equation shown above.

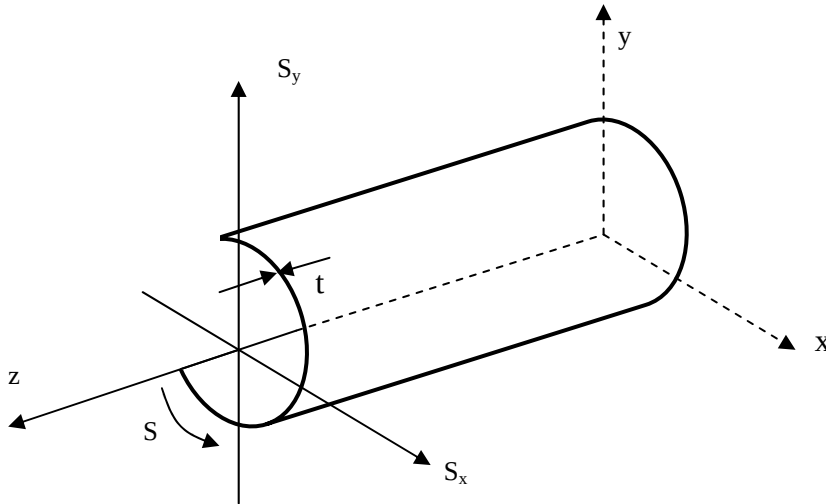
$$\frac{\partial q}{\partial s} + t \frac{\partial \sigma_z}{\partial z} = 0 \quad (A)$$

But from bending theory we have (assuming positive M_y):

$$\sigma_z = \left(\frac{M_y I_{xx} - M_x I_{xy}}{I_{xx} I_{yy} - I_{xy}^2} \right) x + \left(\frac{M_x I_{yy} - M_y I_{xy}}{I_{xx} I_{yy} - I_{xy}^2} \right) y$$

Differentiating σ_z with respect to z and substituting it in the above equation with q :

$$\frac{\partial \sigma_z}{\partial z} = \left(\frac{\left(\frac{\partial M_y}{\partial z} \right) I_{xx} - \left(\frac{\partial M_x}{\partial z} \right) I_{xy}}{I_{xx} I_{yy} - I_{xy}^2} \right) x + \left(\frac{\left(\frac{\partial M_x}{\partial z} \right) I_{yy} - \left(\frac{\partial M_y}{\partial z} \right) I_{xy}}{I_{xx} I_{yy} - I_{xy}^2} \right) y$$



Using the relationship given in stress / displacement function at the beginning where

$$\frac{\partial M_y}{\partial z} = S_x, \text{ The above expression becomes;}$$

$$\frac{\partial \sigma_z}{\partial z} = \left(\frac{S_x I_{xx} - S_y I_{xy}}{I_{xx} I_{yy} - I_{xy}^2} \right) x + \left(\frac{S_y I_{yy} - S_x I_{xy}}{I_{xx} I_{yy} - I_{xy}^2} \right) y$$

Substituting the above in equation (A).

$$\frac{\partial q}{\partial s} = - \left(\frac{S_x I_{xx} - S_y I_{xy}}{I_{xx} I_{yy} - I_{xy}^2} \right) tx - \left(\frac{S_y I_{yy} - S_x I_{xy}}{I_{xx} I_{yy} - I_{xy}^2} \right) ty$$

Integrating the above equation:

$$\int_0^s \frac{\partial q}{\partial s} ds = - \left(\frac{S_x I_{xx} - S_y I_{xy}}{I_{xx} I_{yy} - I_{xy}^2} \right) \int_0^s t x ds - \left(\frac{S_y I_{yy} - S_x I_{xy}}{I_{xx} I_{yy} - I_{xy}^2} \right) \int_0^s t y ds$$

If origin for s is taken at the open edge of the cross section, then $q = 0$ when $s = 0$ and above equation becomes.

$$q_s = - \left(\frac{S_x I_{xx} - S_y I_{xy}}{I_{xx} I_{yy} - I_{xy}^2} \right) \int_0^s t x ds - \left(\frac{S_y I_{yy} - S_x I_{xy}}{I_{xx} I_{yy} - I_{xy}^2} \right) \int_0^s t y ds$$

In the case of unsymmetrical section under the application of S_y only, the above equation reduces to

$$q_s = - \left(\frac{-S_y I_{xy}}{I_{xx} I_{yy} - I_{xy}^2} \right) \int_0^s t x ds - \left(\frac{S_y I_{yy}}{I_{xx} I_{yy} - I_{xy}^2} \right) \int_0^s t y ds$$

For symmetrical section when, $I_{xy} = 0$ the above equation reduces to

$$q_s = - \left(\frac{S_x}{I_{yy}} \right) \int_0^s t x ds - \left(\frac{S_y}{I_{xx}} \right) \int_0^s t y ds$$

And in the absence of say S_y or S_x the above equation further reduces to

Either

$$q_s = - \left(\frac{S_x}{I_{yy}} \right) \int_0^s t x ds$$

Or

$$q_s = - \left(\frac{S_y}{I_{xx}} \right) \int_0^s t y ds$$

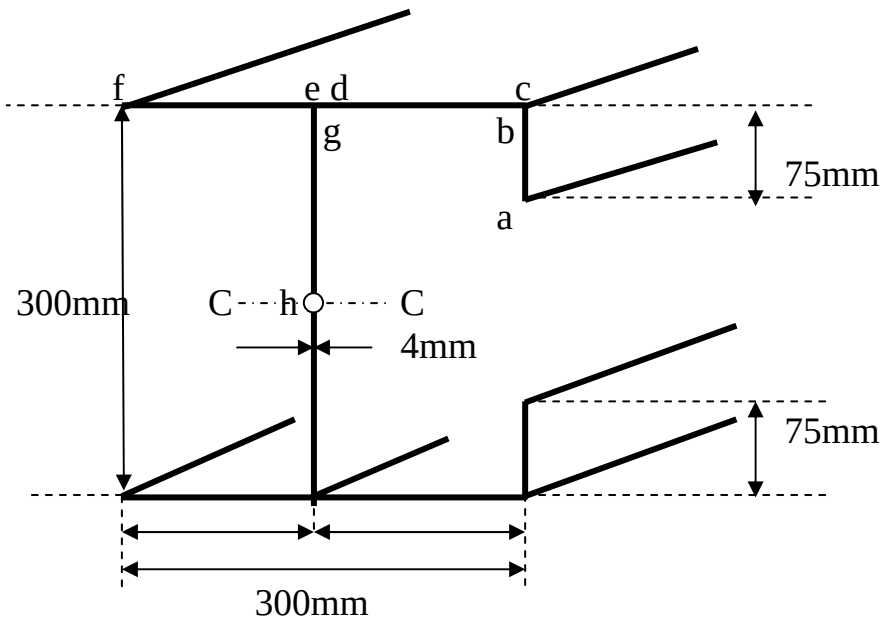
For simple symmetrical section the above equations will take the form of

$$q_s = - \left(\frac{S_y}{I_{xx}} \right) \sum_{i=1}^{i=n} Q_i$$

Where Q is the first moment of area for the segment under consideration about the NA. In symmetrical sections, shear flow distribution is usually linear on the flanges and parabolic along the vertical webs.

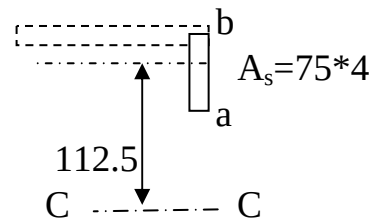
Example 1

Find the shear flow values in the beam cross section shown below in terms of S_y .
For the section shown $I_{cc} = 70.88 \times 10^6 \text{ mm}^4$



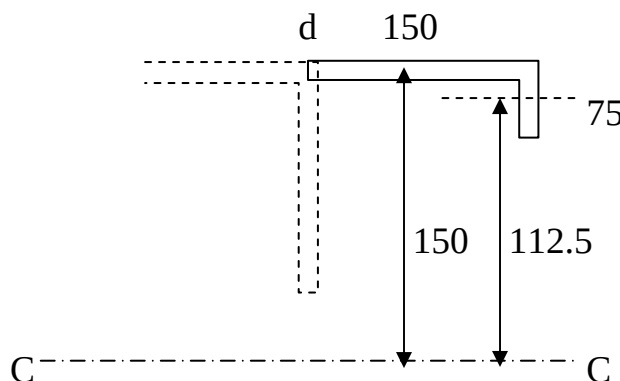
- Basic equation: $q_s = -\left(\frac{S_y}{I_{xx}}\right) \sum_{i=1}^{i=n} Q_i \leftarrow$ of material beyond point considered.

- Usually work to mid-thickness.
- Starting at a 'Free Edge', say at 'a'.
- At 'a' $q = 0$
- At 'b' we have :



$$q_b = \frac{S_y}{70.88 \times 10^6} [112.5 \times (75 \times 4)] = 4.7615 \times 10^{-4} S_y$$

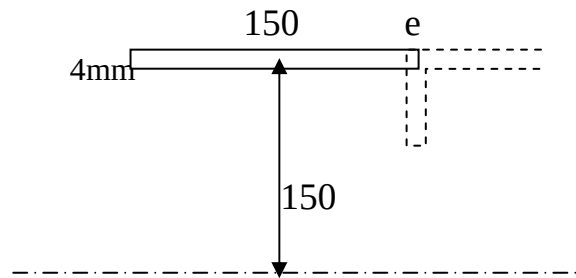
- At 'c' we have : $q_c = q_b$
- At 'd' we have :



$$q_d = \frac{s_y}{70.88 \times 10^6} [112.5(75 \times 4) + 150(150 \times 4)]$$

$$\therefore q_d = 1.745 \times 10^{-3} s_y$$

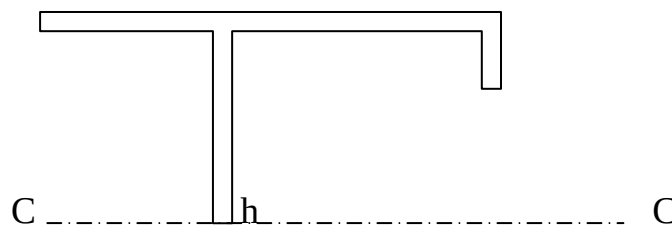
- At 'e' we have :



$$q_e = \frac{s_y}{70.88 \times 10^6} [150(150 \times 4)] = 1.2697 \times 10^{-3} s_y$$

- At 'f' we have : $q_f = 0$ (Free edge)
- At 'g' we have : $q_g = q_d + q_e$
 $q_g = 3.0147 \times 10^{-3} S_y$

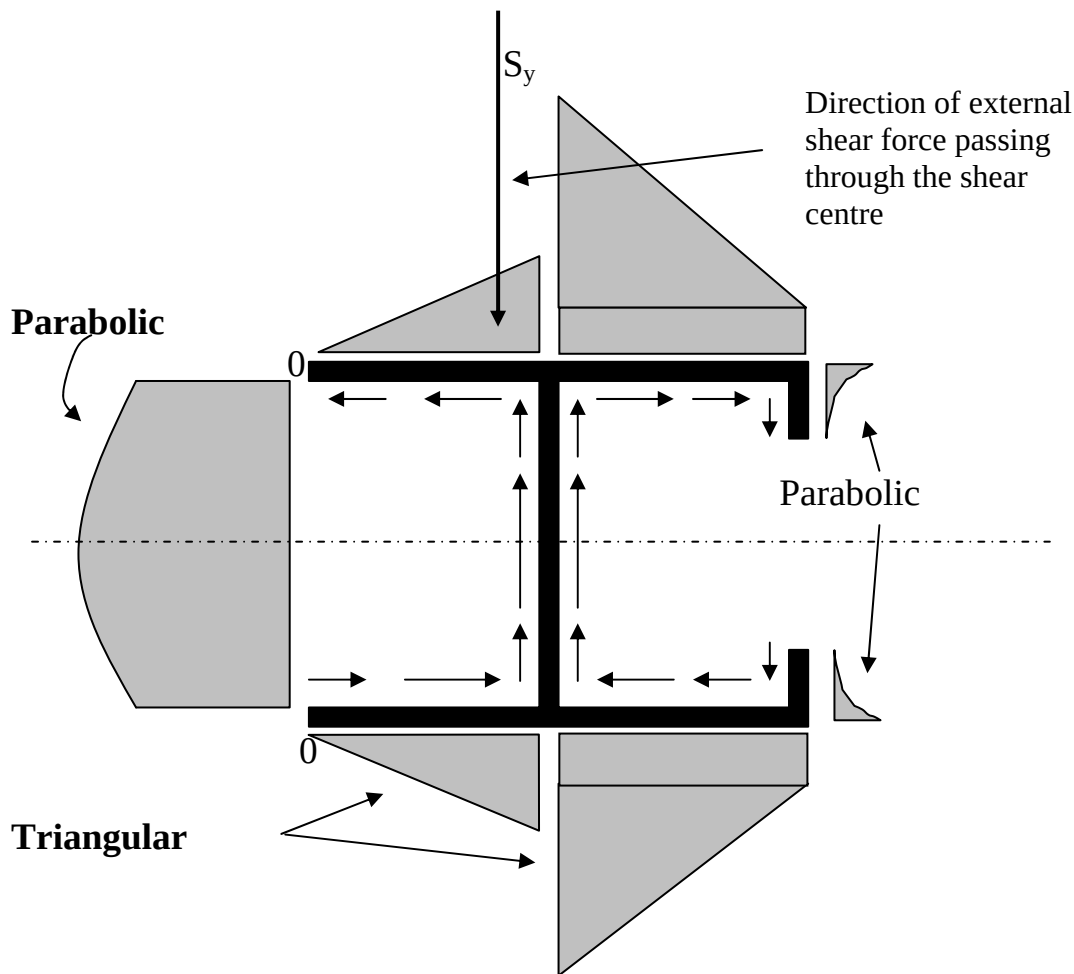
- At 'h' we have :



$$q_h = q_g + \frac{s_y}{70.88 \times 10^6} [75(150 \times 4)]$$

$$\therefore q_h = 3.6496 \times 10^{-3} s_y$$

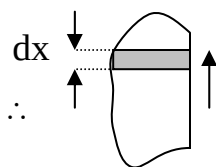
- We can now plot the shear flow distribution :



SHEAR FLOW DISTRIBUTION

NOTE: Direction of arrows in main vertical web opposes applied force, all other arrows flow in to or away from these!

Force:



Consider a segment of the above Diagram.

Force over length $dx = q dx = (\text{shear flow} \times \text{distance})$

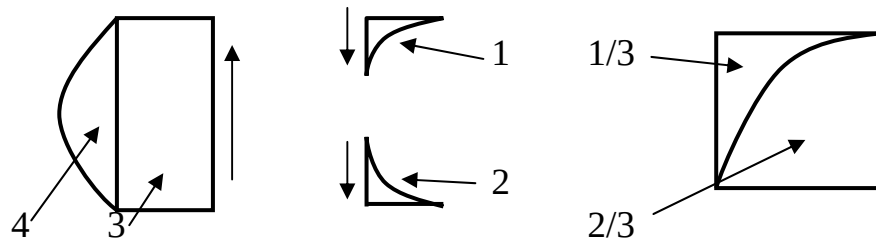
Total force over length, $l = \int_0^l q dx$

Force = Area of 'q' diagram.

Load Balance

One use of the concept Force = Area of 'S' diagram, is that it enable us a check for 'Load Balance' to be done. In the example above the applied shear force was ' S_y ' applied vertically. There are 3 sets of vertical 'S' arrows: -

NB. For parabola area=2/3 or 1/3 of the surrounding rectangle



So, Areas;

$$1 = \frac{1}{3} \times 75 \times 4.7615 \times 10^{-4} S_y \downarrow$$

$$2 = \frac{1}{3} \times 75 \times 4.7615 \times 10^{-4} S_y \downarrow$$

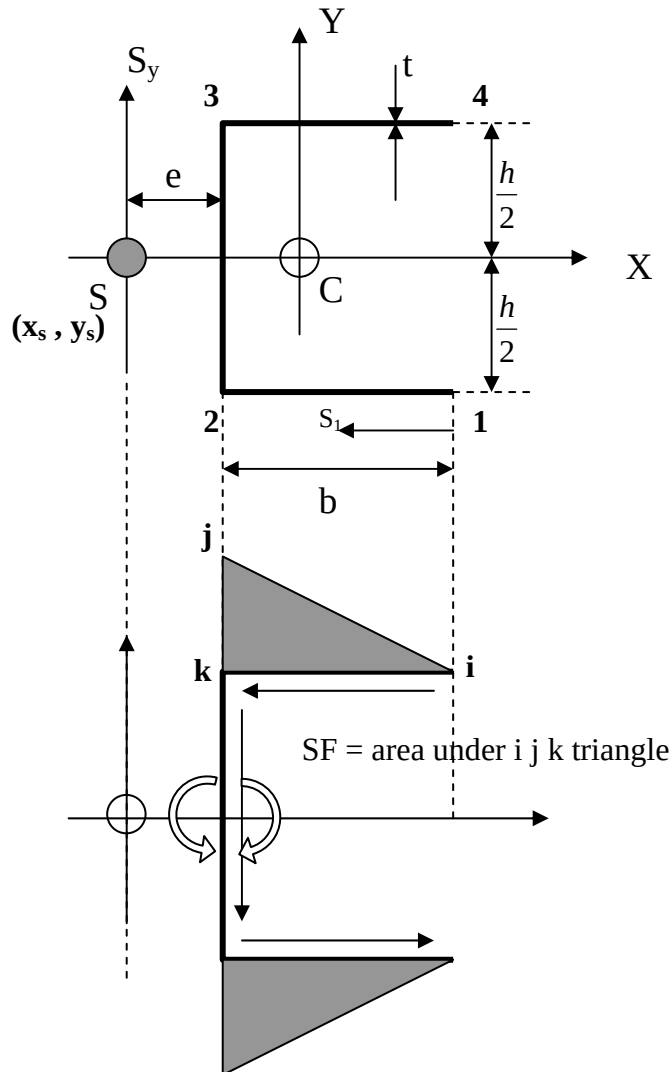
$$3 = 300 \times 3.0147 \times 10^{-3} S_y \uparrow$$

$$4 = \frac{2}{3} \times 300 \times 6.349 \times 10^{-4} S_y \uparrow$$

Sum = 1.00758 $\therefore$ 0.7 % error!

Shear Centre

Shear centre is defined as the point in the cross section through which shear loads produce no twisting. This is also the centre of twist of the section subject to pure torsion. In the case of symmetrical sections this point lies on the axis of symmetry.



For the channel section shown above, Cx is an axis of symmetry so $I_{xy} = 0$.

Also $S_x = 0$.

Therefore the general equation shown previously reduces to

$$q_s = -\left(\frac{S_y}{I_{xx}}\right) \int_0^s t y ds$$

where $I_{xx} = 2bt\left(\frac{h}{2}\right)^2 + \frac{th^3}{12} = \frac{t}{12}h^3\left(1 + \frac{6b}{h}\right)$

Substituting this for I_{xx} in the above equation we have:

$$q_s = \left(\frac{-12S_y}{h^3(1 + 6b/h)} \right) \int_0^s y ds$$

On the bottom flange, $y = -\frac{h}{2}$ so that we have

$$q_{12} = \left(\frac{6S_y}{h^2(1 + 6b/h)} \right) s_1$$

Equating the counter clockwise moments of the internal shears about the mid-point of the web to the clockwise moment of the applied shear load through the shear centre, about the same point gives:

$$S_y \cdot e = 2 \int_0^b q_{12} \frac{h}{2} ds_1$$

$$S_y \cdot e = 2 \int_0^b \left(\frac{6S_y}{h^2(1 + 6b/h)} \right) \frac{h}{2} s_1 ds_1$$

From which:

$$e = \frac{2 \int_0^b q_{12} \frac{h}{2} ds_1}{S_y} = \frac{3b^2}{h(1 + 6b/h)}$$

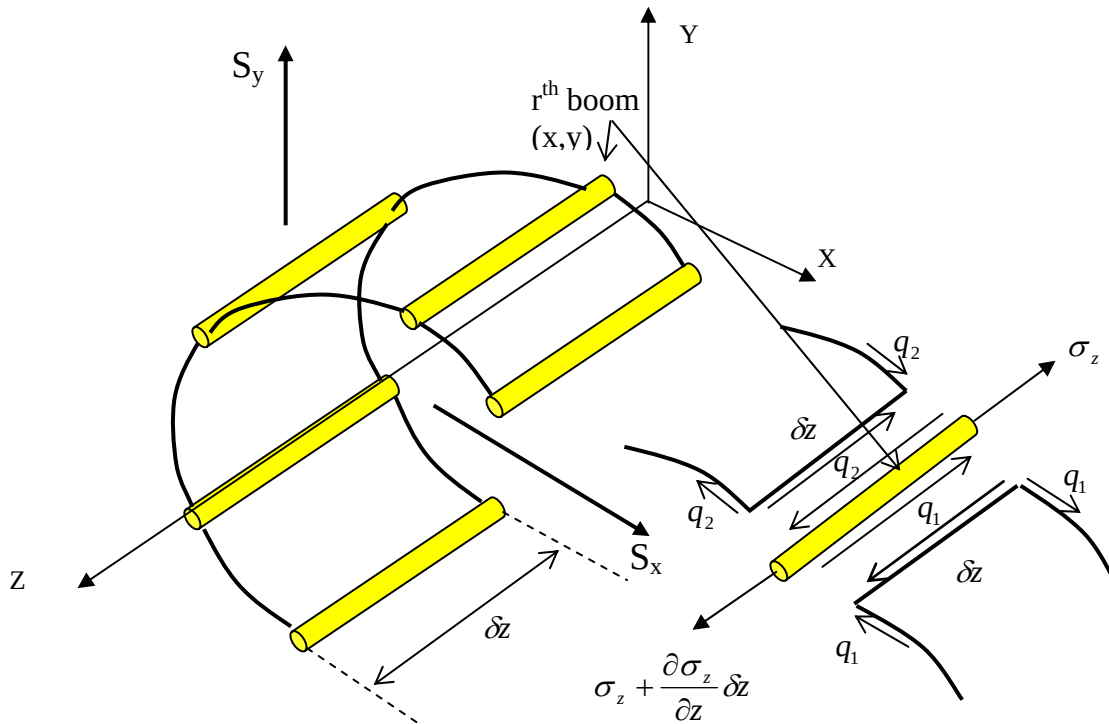
Very Important Note:-

The area under the shear flow diagram is equivalent to internal shear force.
Therefore the force on the top flange is equal to the area of the triangle *ijk*.

Effects of idealization on the shear flow analysis of open and closed section beams.

Above modification, i.e. the introduction of direct stress carrying booms clearly modifies the analysis of open and closed section beams.

Effects of idealisation on shear flow analysis



The equation below is for shear flow distribution in the cross section of an open section beam. It is based on the equilibrium shown previously where t is the thickness of the direct stress carrying skin, referred to as t_D .

I.e. $t_D = t$, if skin is carrying direct stress or $t_D = 0$ if the skin is assumed to carry shear stresses only.

$$q_s = - \left(\frac{S_x I_{xx} - S_y I_{xy}}{I_{xx} I_{yy} - I_{xy}^2} \right) \int_0^s t_D x ds - \left(\frac{S_y I_{yy} - S_x I_{xy}}{I_{xx} I_{yy} - I_{xy}^2} \right) \int_0^s t_D y ds$$

This equation does not take into account the effect of booms, which produces shear stress discontinuities, which interrupts shear flow. To modify the above equation to cater for idealised section we consider the equilibrium of the r^{th} boom in the elemental length of the beam which carries shear loads S_x and S_y acting through the

shear centre C, shown above. These shear stresses produce direct stress due to bending in the boom and skin and shear stress in the skin.

Suppose the shear flows in the skin adjacent to a boom with cross section area B_r are q_1 and q_2 . Then, from above diagram

$$\left(\sigma_z + \frac{\partial \sigma_z}{\partial z} \delta z \right) B_r - \sigma_z B_r + q_2 \delta z - q_1 \delta z = 0$$

Which simplifies to

$$q_2 - q_1 = - \frac{\partial \sigma_z}{\partial z} B_r$$

Using the equation for σ_z from the bending theory, in the above equation and differentiating it w.r.t Z, we get:

$$q_2 - q_1 = - \left(\frac{\left(\frac{\partial M_y}{\partial z} \right) I_{xx} - \left(\frac{\partial M_x}{\partial z} \right) I_{xy}}{I_{xx} I_{yy} - I_{xy}^2} \right) B_r x_r - \left(\frac{\left(\frac{\partial M_x}{\partial z} \right) I_{yy} - \left(\frac{\partial M_y}{\partial z} \right) I_{xy}}{I_{xx} I_{yy} - I_{xy}^2} \right) B_r y_r$$

Or

$$q_2 - q_1 = - \left(\frac{S_x I_{xx} - S_y I_{xy}}{I_{xx} I_{yy} - I_{xy}^2} \right) B_r x_r - \left(\frac{S_y I_{yy} - S_x I_{xy}}{I_{xx} I_{yy} - I_{xy}^2} \right) B_r y_r$$

The above equation gives the changes in shear flow induced by boom which itself is subjected to direct load of $\sigma_z B_r$. Each time a boom is encountered the shear flow is incremented by this amount so that if at any distance S around the profile of the section, n booms have passed, the shear flow at that point is given by:

$$q_s = - \left(\frac{S_x I_{xx} - S_y I_{xy}}{I_{xx} I_{yy} - I_{xy}^2} \right) \left(\int_0^s t_D x ds + \sum_{r=1}^n B_r x_r \right) - \left(\frac{S_y I_{yy} - S_x I_{xy}}{I_{xx} I_{yy} - I_{xy}^2} \right) \left(\int_0^s t_D y ds + \sum_{r=1}^n B_r y_r \right)$$

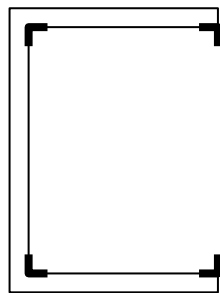
This general expression when applied to a symmetrical section (say about x), with S_y acting alone and $t_D = 0$, reduces to:

$$q_s = -\left(\frac{S_y}{I_{xx}}\right)\left(\sum_{r=1}^n B_r y_r\right)$$

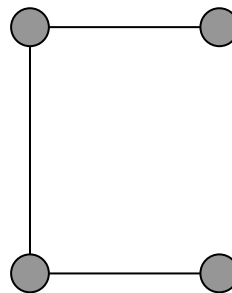
The shear flow distribution along and across the panels will remain constant since

$$t_D = 0.$$

Example: -



Idealised to



Example 2

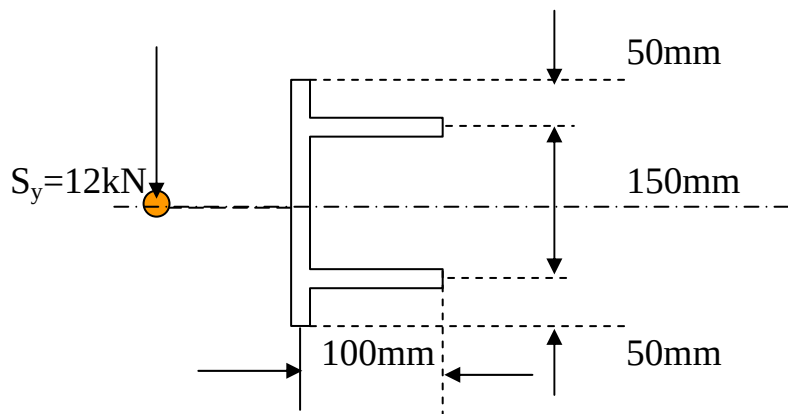
Determine the location of the shear centre O, and the distribution of the shear flows and stresses caused by a vertical shear force S applied through the shear centre.

Assume:

$$I_{xx} = 7.28 \text{E}6 \text{mm}^4,$$

$$S_y = 12 \text{kN}$$

3mm thickness throughout.



We use $q_s = -\left(\frac{S_y}{I_{xx}}\right)\sum Q$

The idealised section is shown below

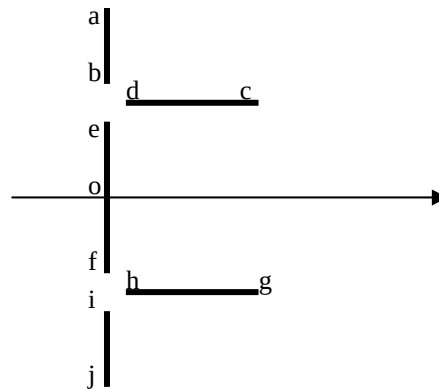
$$q_a = 0$$

$$q_b = \frac{S_y}{I_{xx}} Q_{ab}$$

$$q_c = 0$$

$$q_d = \frac{S_y}{I_{xx}} Q_{cd}$$

$$q_e = q_d + q_b = \frac{S_y}{I_{xx}} Q_{ab} + \frac{S_y}{I_{xx}} Q_{cd} = \frac{S_y}{I_{xx}} (Q_{ab} + Q_{cd})$$



$$q_o = q_e + \frac{S_y}{I_{xx}} Q_{eo} = q_d + q_b + \frac{S_y}{I_{xx}} Q_{eo} = \frac{S_y}{I_{xx}} Q_{ab} + \frac{S_y}{I_{xx}} Q_{cd} + \frac{S_y}{I_{xx}} Q_{eo} = \frac{S_y}{I_{xx}} (Q_{ab} + Q_{cd} + Q_{eo})$$

Where:

Q_{ij} = the first moment of area of segment between I and j .

I_{xx} is the first moment of area of the section about xx.

$$I_{xx} = \sum \frac{bd^3}{12} + Ah^2 = 2 \left[\left(\frac{100 * 3^3}{12} \right) + 100 * 3 * 75^2 \right] + \frac{3 * 250^3}{12}$$

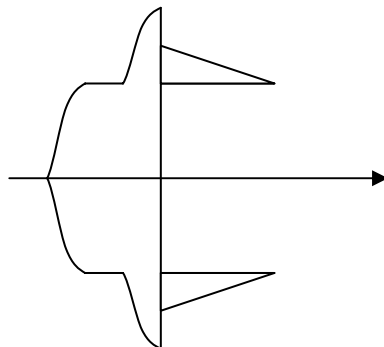
And

$$Q_{ab} = 3 * 50 * 100$$

$$Q_{cd} = 3 * 100 * 75$$

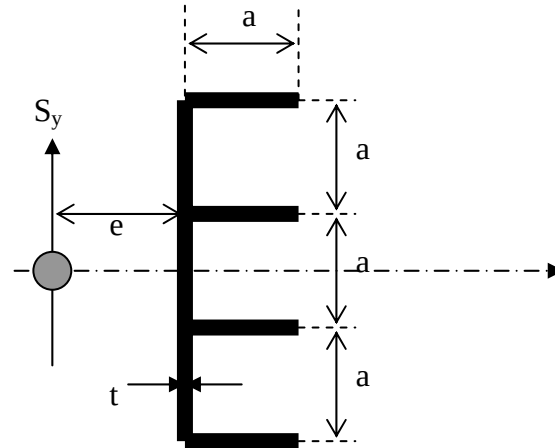
$$Q_{ef} = 3 * 75 * 37.5$$

Due to symmetry we do not need to calculate the shear flows on the lower half.
The lower half is the mirror image of the top half.



Example 3

Determine the shear flow distribution in a thin walled Z-section, shown below, due to shear load S_y applied through the shear centre of the section. Determine the location of the shear centre. Assume $a=30\text{mm}$, $t=4\text{mm}$ and $S_y = 12\text{kN}$

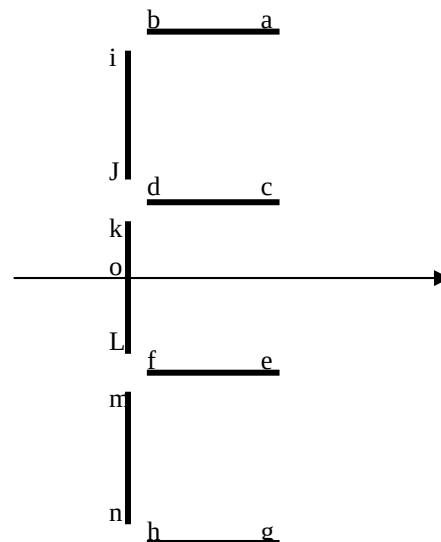


Solution

Due to symmetry and because $S_x = 0$

We use $q_s = -\left(\frac{S_y}{I_{xx}}\right) \sum Q$

The idealised section is shown here



$$q_a = 0$$

$$q_b = \frac{S_y}{I_{xx}} Q_{ab}$$

$$q_c = 0$$

$$q_d = \frac{S_y}{I_{xx}} Q_{cd}$$

$$q_i = q_b$$

$$q_j = q_i + \frac{S_y}{I_{xx}} Q_{ij} = \frac{S_y}{I_{xx}} Q_{ab} + \frac{S_y}{I_{xx}} Q_{ij}$$

$$q_k = q_j + q_d = \frac{S_y}{I_{xx}} Q_{ab} + \frac{S_y}{I_{xx}} Q_{ij} + \frac{S_y}{I_{xx}} Q_{cd}$$

$$q_o = q_k + \frac{S_y}{I_{xx}} Q_{ko} = \frac{S_y}{I_{xx}} Q_{ab} + \frac{S_y}{I_{xx}} Q_{ij} + \frac{S_y}{I_{xx}} Q_{cd} + \frac{S_y}{I_{xx}} Q_{ko}$$

$$q_o = \frac{S_y}{I_{xx}} (Q_{ab} + Q_{ij} + Q_{cd} + Q_{ko})$$

Where:

Q_{ij} = the first moment of area of segment between i and j .

I_{xx} is the first moment of area of the section about xx .

$$I_{xx} = \sum \frac{bd^3}{12} + Ah^2 =$$

$$2 \left[\left(\frac{a * t^3}{12} \right) + a * t * \left(\frac{3}{2} a \right)^2 \right] + \frac{t * (3a)^3}{12} + 2 \left[\left(\frac{a * t^3}{12} \right) + a * t * \left(\frac{1}{2} a \right)^2 \right]$$

And

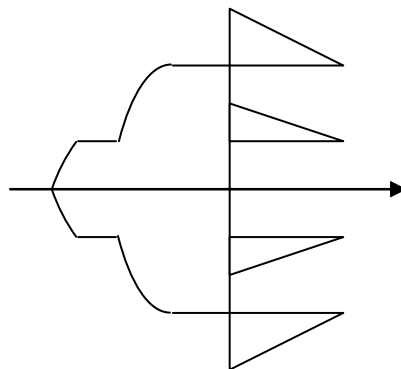
$$Q_{ab} = t * a * (3/2) * a$$

$$Q_{cd} = t * a * (1/2) * a$$

$$Q_{ij} = t * a * a$$

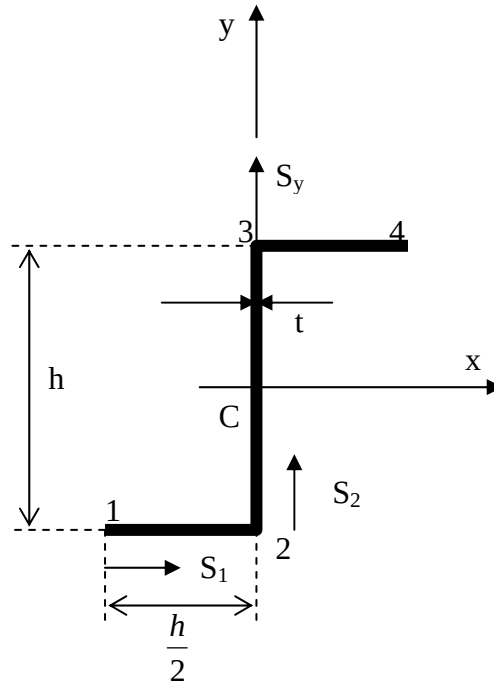
$$Q_{ko} = t * a * a / 4$$

Due to symmetry we do not need to calculate the shear flows on the lower half.
The lower half is the mirror image of the top half.



Example 4

Determine the shear flow distribution in a thin walled Z-section, shown below, due to shear load S_y applied through the shear centre of the section



Since $S_x = 0$ and $I_{xy} \neq 0$, the general equation reduces to

$$q_s = - \left(\frac{-S_y I_{xy}}{I_{xx} I_{yy} - I_{xy}^2} \right) \int_0^s t x ds - \left(\frac{S_y I_{yy}}{I_{xx} I_{yy} - I_{xy}^2} \right) \int_0^s t y ds$$

$$q_s = \frac{S_y}{I_{xx} I_{yy} - I_{xy}^2} \left(I_{xy} \int_0^s t x ds - I_{yy} \int_0^s t y ds \right)$$

assuming that

$$I_{xx} = \frac{h^3 t}{3}, \quad I_{yy} = \frac{h^3 t}{12}, \quad I_{xy} = \frac{h^3 t}{8}$$

Substituting all of the above in the general equation shown above:

$$q_s = \frac{S_y}{h^3} \int_0^s (10.32x - 6.84y) ds$$

At the bottom flange 1-2, $y = -\frac{h}{2}$ and $x = -\frac{h}{2} + s_1$ where $0 \leq s_1 \leq \frac{h}{2}$ therefore

$$q_{12} = \frac{S_y}{h^3} \int_0^s (10.32s_1 - 1.74h) ds_1$$

giving

$$q_{12} = \frac{S_y}{h^3} (5.16 s_1^2 - 1.74 h s_1)$$

Hence at 1 $s_1 = 0$, $q_1 = 0$

$$\text{at 2 } s_1 = \frac{h}{2} \quad q_2 = \frac{0.42 S_y}{h}$$

Further examination of the above equation shows that shear from distribution is parabolic with change of sign at $s_1 = 0.336h$.

In the web 2-3, $y = -\frac{h}{2} + s_2$ and $x = 0$ where $0 \leq s_2 \leq h$ therefore

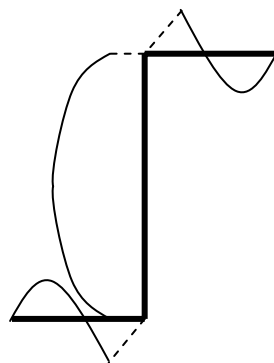
$$q_{23} = \frac{S_y}{h^3} \int_0^{s_2} (3.42 h - 6.84 s_2) ds_2 + q_2$$

above equation is not zero at $s_2 = 0$, it is equal to q_2 calculated previously.

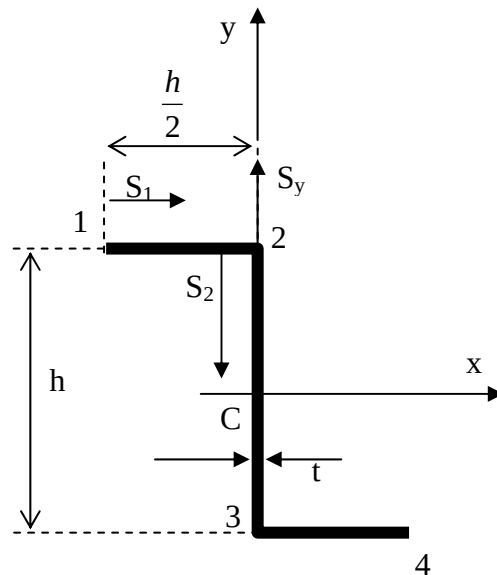
$$q_{23} = \frac{S_y}{h^3} (0.42 h^2 + 3.42 h s_2 - 3.42 s_2^2)$$

The distribution is symmetrical about Cx, with maximum at $s_2 = \frac{h}{2}$ or $y=0$

Shear flow at the top flange is antisymmetric, therefore it can be deduced from the shear flow in the lower flange.



Determine the shear flow distribution in a thin walled Z-section, shown below, due to shear load S_y applied through the shear centre of the section



Since $S_x = 0$ and $I_{xy} \neq 0$, the general equation reduces to

$$q_s = - \left(\frac{-S_y I_{xy}}{I_{xx} I_{yy} - I_{xy}^2} \right) \int_0^s t x ds - \left(\frac{S_y I_{yy}}{I_{xx} I_{yy} - I_{xy}^2} \right) \int_0^s t y ds$$

$$q_s = \frac{S_y}{I_{xx} I_{yy} - I_{xy}^2} \left(I_{xy} \int_0^s t x ds - I_{yy} \int_0^s t y ds \right)$$

assuming that

$$I_{xx} = \frac{h^3 t}{3}, \quad I_{yy} = \frac{h^3 t}{12}, \quad I_{xy} = -\frac{h^3 t}{8}$$

Substituting all of the above in the general equation shown above:

$$q_s = \frac{S_y}{h^3} \int_0^s (-10.32x - 6.84y) ds$$

At the top flange 1-2, $y = +\frac{h}{2}$ and $x = -\frac{h}{2} + s_1$ where $0 \leq s_1 \leq \frac{h}{2}$ therefore

$$q_{12} = \frac{S_y}{h^3} \int_0^{s_1} (-10.32s_1 + 1.74h) ds_1$$

giving

$$q_{12} = \frac{S_y}{h^3} (-5.16 s_1^2 + 1.74 h s_1)$$

Hence at 1 $s_1 = 0$, $q_1 = 0$

$$\text{at 2 } s_1 = \frac{h}{2} \quad q_2 = -\frac{0.42h^2 S_y}{h^3} = -\frac{0.42 S_y}{h}$$

Further examination of the above equation shows that shear from distribution is parabolic with change of sign at $s_1 = 0.336h$.

In the web 2-3, $y = +\frac{h}{2} - s_2$ and $x = 0$ where $0 \leq s_2 \leq h$ therefore

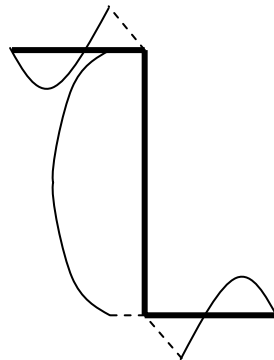
$$q_{23} = \frac{S_y}{h^3} \int_0^{s_2} (-3.42h + 6.84s_2) ds_2 + q_2$$

above equation is not zero at $s_2 = 0$, it is equal to q_2 calculated previously.

$$q_{23} = \frac{S_y}{h^3} (-0.42h^2 - 3.42hs_2 + 3.42s_2^2)$$

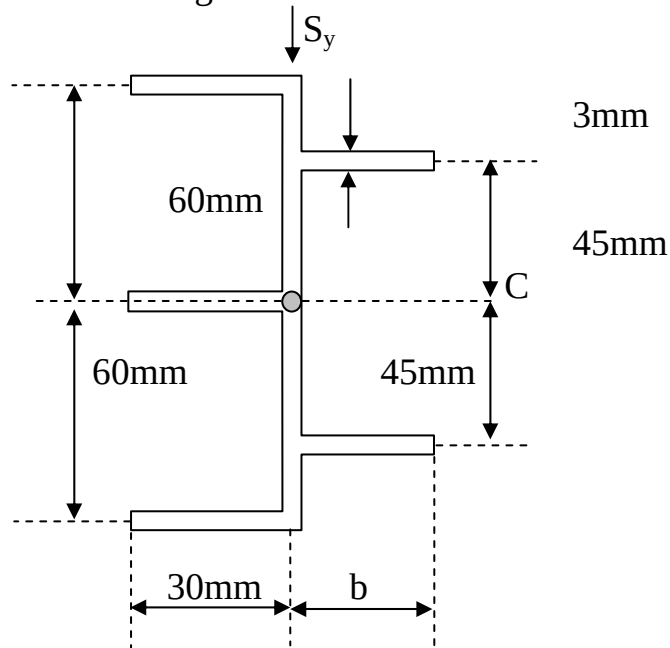
The distribution is symmetrical about Cx, with maximum at $s_2 = \frac{h}{2}$ or $y=0$

Shear flow at the top flange is antisymmetric, therefore it can be deduced from the shear flow in the lower flange.

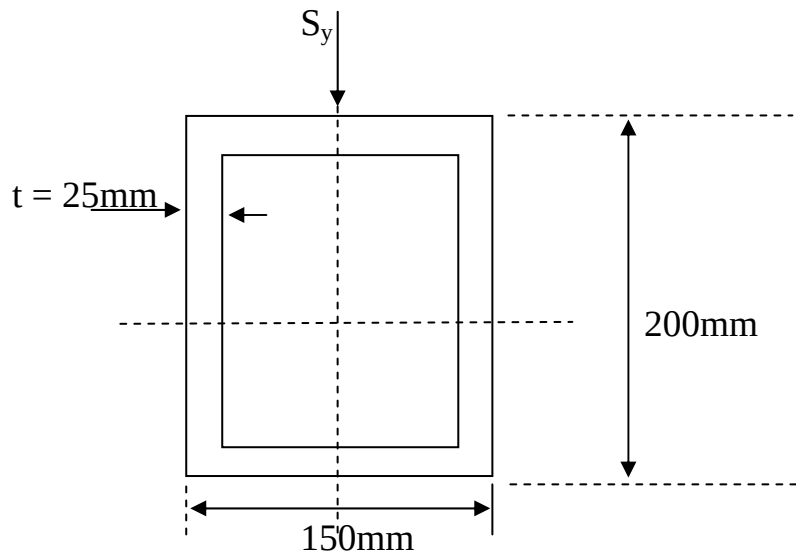


Tutorials

- 1) A Thin walled beam of uniform thickness 3mm has the cross section shown below. Determine;
 - a) The dimension b for which the shear centre C of the beam is located at the point indicated.
 - b) Sketch the shear stress distribution on the section when it is subjected to a vertical shear force V through the shear centre as shown below.



- 2) The thin-walled box section beam shown below is subjected to a shear force of 20kN.



Determine:

- Shear stress at A
- Shear stress at B
- Shear stress at C
- Sketch the shear stress distributions
- Explain what is meant by shear centre and its significance and position in this question.

- 3) Discuss shear flow and its relation to shear force and shear stress in a section.

For the extruded beam having the cross section shown below.

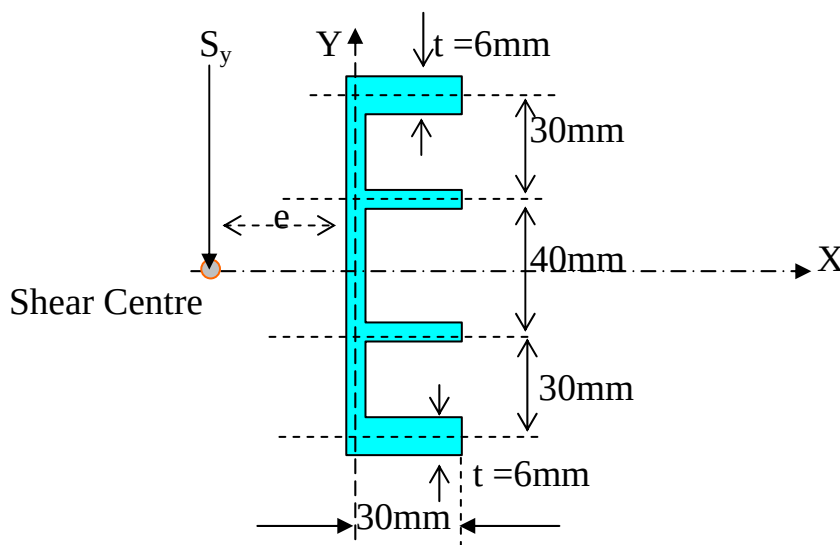
- b) Determine the location of the shear centre.
a) Sketch the distribution of the shearing stresses in the section due to the application of vertical shearing force S_y applied at the shear centre, as shown.

Assume;

$$I_x = 1.149 \text{E}6 \text{mm}^4,$$

$$S_y = 1 \text{kN}$$

The thickness is 4mm every where unless stated otherwise.



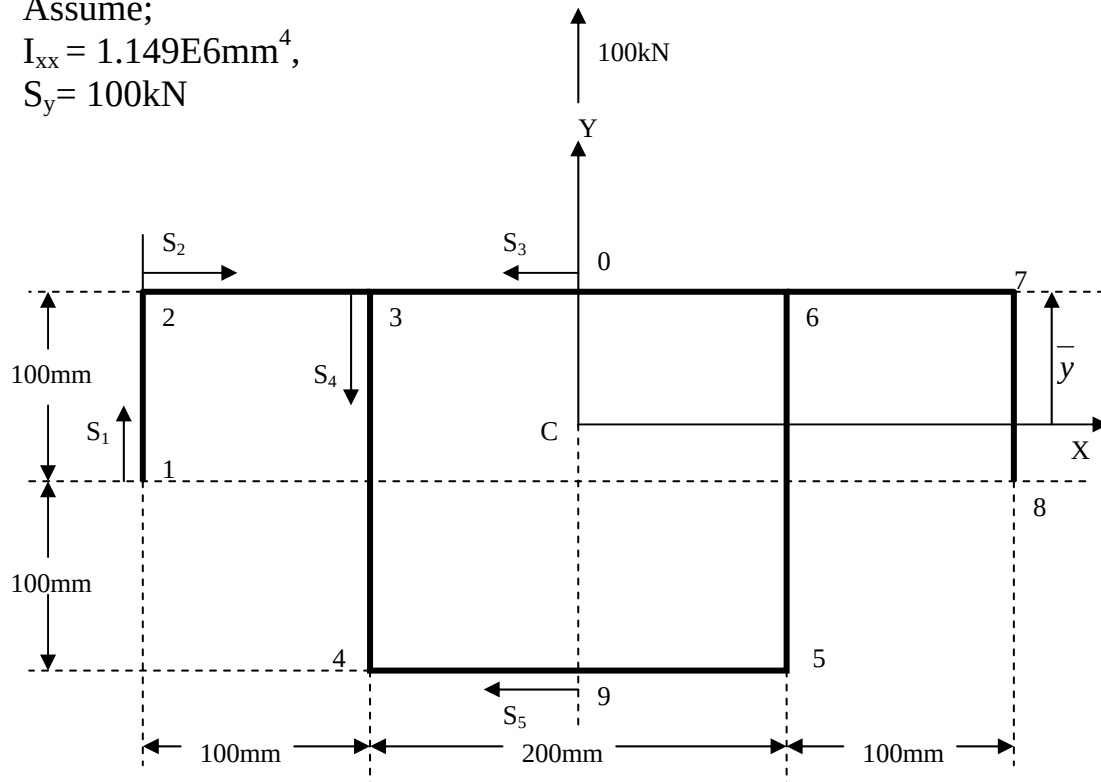
4) For the extruded beam having the cross section shown below.

- c) Determine the location of the shear centre.
- b) Sketch the distribution of the shearing stresses in the section due to the application of vertical shearing force S_y applied at the shear centre, as shown.

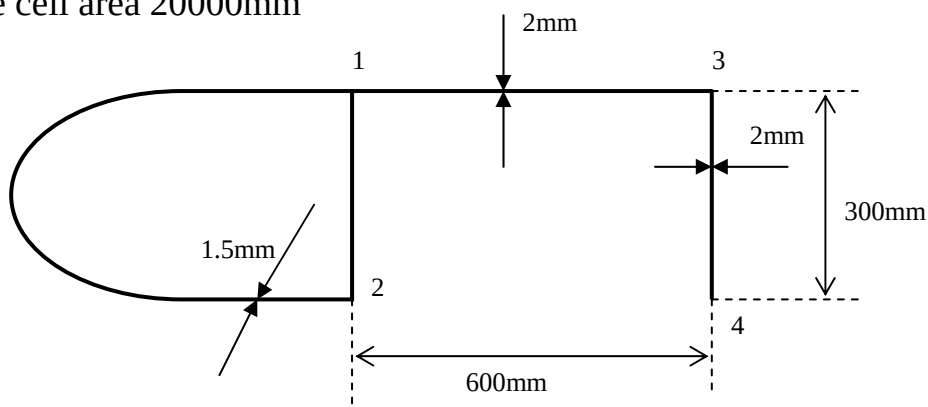
Assume;

$$I_{xx} = 1.149E6 \text{mm}^4,$$

$$S_y = 100 \text{kN}$$



Nose cell area 20000mm^2



Torsion

When torque is applied to a component it twists and the mode of deformation is called Torsion. Torsion can be applied to all solid or hollow, circular or non-circular members. Torsion in components usually results in an angle of twist as well as shear stress in the wall of the component. The angle of twist and average shearing stresses are related to other parameters that govern TORSION by the Engineering Torsion Formula, shown below.

$$\frac{T}{J} = \frac{\tau}{r} = \frac{G\theta}{L}$$

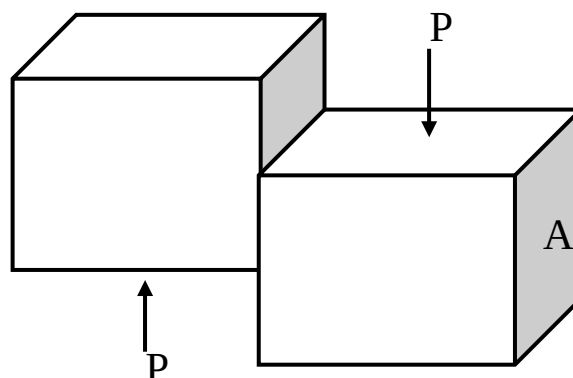
This lecture concentrates solely on the torsion of the circular, hollow and solid circular shafts. However non-circular shafts can also experience torsion but the associated theory is more complicated and will be discussed at a later stage of this module.

This lecture is about:

- the derivation of the above equality
- the definition of its individual terms
- the assumptions made
- the limitations put on its application

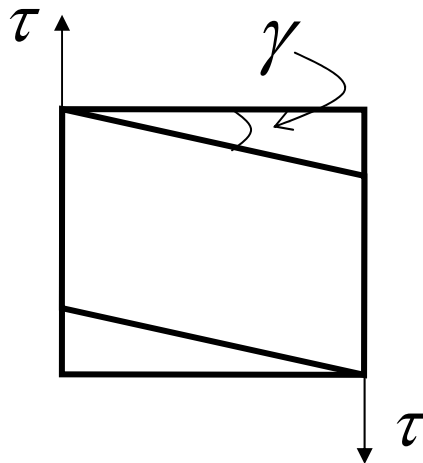
The definitions

a) Average Shearing Stress, τ

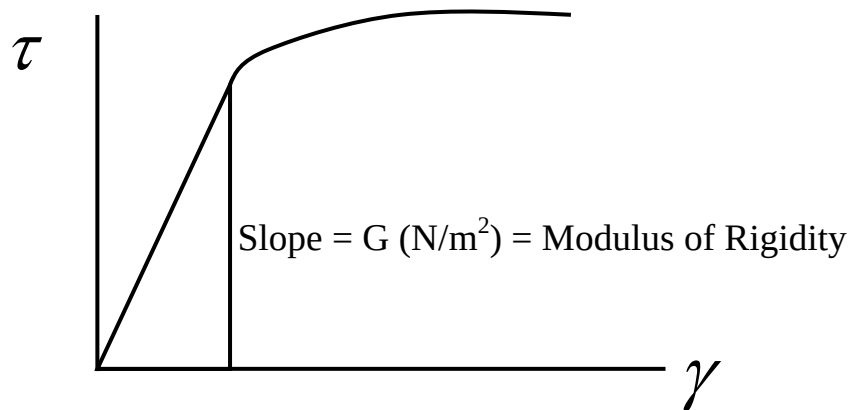


$$\tau = \frac{P}{A}$$

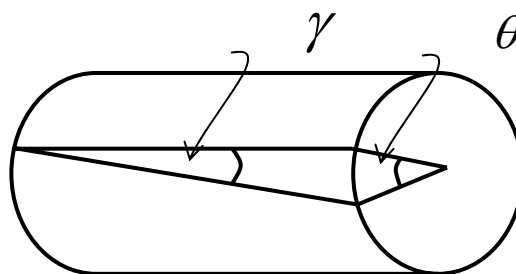
(b) Shearing Strain γ (Rads)



(c) Shear Modulus, G (N/m^2)

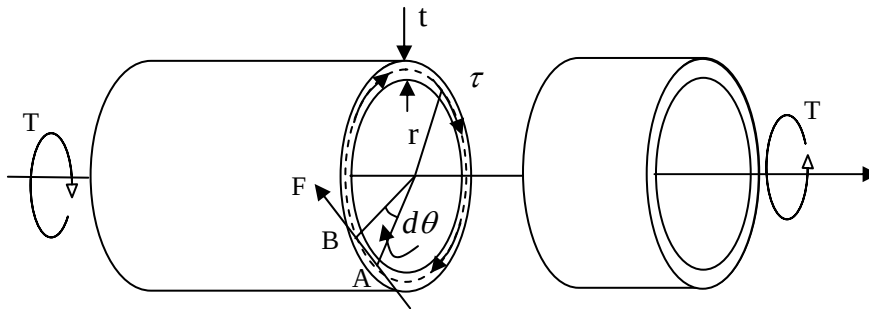


(d) Angle of Twist θ



Torsion of thin-walled cylinders

Consider the tube shown below:



Assume the tube;

- is subjected to two opposite torques, T
- has a constant thickness t throughout
- has a length L
- has mean radius r
- has constant average shear stress on its median line

In this case the deformation will be twisting.

As a result all internal sections experience shear stress τ ,

At any section along the length of the tube the internal reaction must be equal to the external torque.

The force F due to shear stress τ acting on an incremental element AB along a section is given by:

$$F = \tau \times r \times d\theta \times t$$

The moment ΔT of this force about the centre line is.

$$\Delta T = F \times r$$

The total sum of all the internal moments of resistance (reaction) must be equal to the externally applied torque causing the twist.

$$T = \sum \Delta T = \int_0^{2\pi} \tau \times r^2 \times t \times d\theta = 2\pi \cdot r^2 \cdot t \cdot \tau = 2A \cdot t \tau$$

Where, $A = \pi \cdot r^2$ is the area of the circle bounded by the median line.

$$\tau = \frac{T}{2At}$$

Or in terms of shear flow q , which is the shear force per unit length

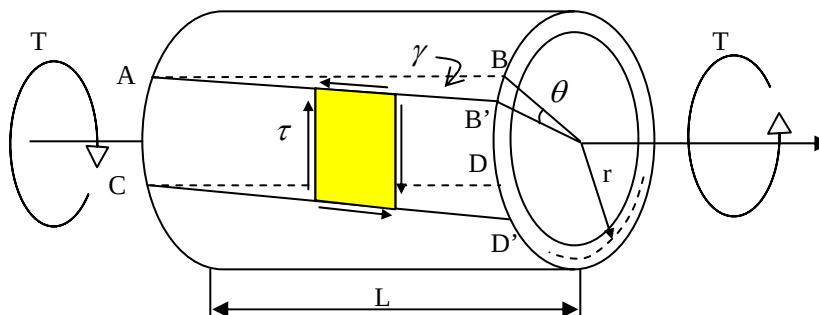
$$q = \frac{T}{2A}$$

Where $q = \tau \cdot t$

Note that this is a very important relationship and can be applied to both circular and non-circular sections. However, in this study we apply it to circular sections only.

Relationship between τ and θ

The rotation of one end of the shaft through an angle θ relative to the other end results in the generation of γ as shown below.



The angle γ is the shear strain, based on the above definition.

From the diagram above the arc BB' can be expressed as:

$$BB' = r\theta = L\gamma$$

From which we can calculate;

$$\gamma = \frac{r\theta}{L}$$

Using shear stress strain relationship, given by Hooke's Law which is;

$$\tau = G\gamma$$

Which results in:

$$\tau = \frac{Gr\theta}{L} = \frac{T}{2At} = \tau = \frac{T}{2\pi r^2 t}$$

or

$$\boxed{\frac{\tau}{r} = \frac{G\theta}{L}}$$

Torsion of solid circular shafts

For solid circular shafts the shear stress varies over the cross section.

Assumptions made are:

- The shaft is straight with uniform cross section over its entire length.
- Torque is constant along the length of the shaft
- Plane cross sections remain plane during twisting
- Radial lines remain radial during twisting
- Deformation is due to the rotation of one plane relative to another
- Planes remain normal to the axis of the shaft

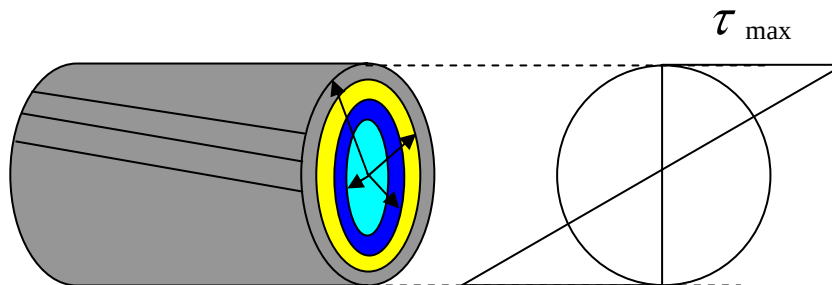
By similar analogy to that of hollow tube and by assuming that a solid shaft is built from a number of concentric hollow tubes each having inner and outer radii r_i and r_o , we have;

$$\frac{\gamma_i L}{r_i} = \frac{\gamma_o L}{r_o} = \theta \quad \text{or} \quad \frac{\gamma_i}{r_i} = \frac{\gamma_o}{r_o} = \text{Constant}$$

Which shows when $r = 0$, $\gamma = 0$ and at the outer surface γ is also maximum resulting in maximum shearing stress being at the outer skin and the variation is linear.

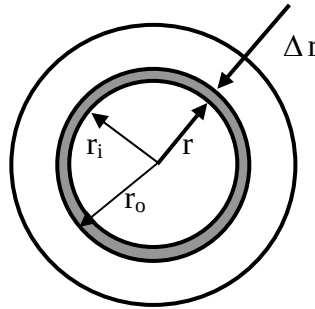
Therefore:

$$\frac{\tau}{r} = \frac{G\theta}{L}$$



Relationship between Torque and Shear Stress.

Consider the equilibrium of one of the thin tubes of inner radius r and thickness Δr on which the average shear stress is τ .



In terms of shear stress, the force per unit length of the circumference is;

$$\delta F = \tau \cdot 1 \cdot \delta r$$

The torque per unit length, associated with this force, about the centre of the shaft is;

$$\delta T = \delta F \cdot r = \tau \cdot r \cdot \delta r$$

The total resisting torque on the whole tube is

$$\Delta T = 2\pi r \cdot \tau \cdot r \cdot \delta r$$

Integrating ΔT over the whole cross section results into the internal moment of resistance, which must be equal to the externally applied torque T , required to maintain equilibrium.

$$T = \int_{r_i}^{r_o} 2\pi r^2 \cdot \tau \cdot \delta r$$

Replacing for $\tau = \frac{Gr\theta}{L}$ we get:

$$T = \int_{r_i}^{r_o} 2\pi r^2 \frac{Gr\theta}{L} \delta r$$

$$T = \frac{G\theta}{L} \int_{r_i}^{r_o} 2\pi r^3 \delta r$$

The integral function:

$$\int_{r_i}^{r_o} 2\pi r^3 \delta r = \frac{\pi}{2} (r_o^4 - r_i^4)$$

Above is known as the Polar Second Moment of Area usually denoted as ***J***

$$\therefore J = \frac{\pi}{2} (r_o^4 - r_i^4) \quad \text{or} \quad J = \frac{\pi}{32} (d_o^4 - d_i^4)$$

Where *d* is the diameter

For a solid circular shaft where $r_i = 0$ the above equation takes the form of

$$J = \int_0^r 2\pi r^3 \delta r = \frac{\pi}{2} (r^4) \quad \text{or} \quad \frac{\pi}{32} (d^4)$$

Therefore we can write

$$\boxed{\frac{T}{J} = \frac{G\theta}{L}}$$

Combining this with $\frac{\tau}{r} = \frac{G\theta}{L}$ from before we have:

$$\frac{\tau}{r} = \frac{G\theta}{L} = \frac{T}{J}$$

Where:-

T = Torque [Nm]

J = Polar Second Moment Of Area [m⁴]

G = Modulus of Rigidity, or Shear Modulus [N/m²]

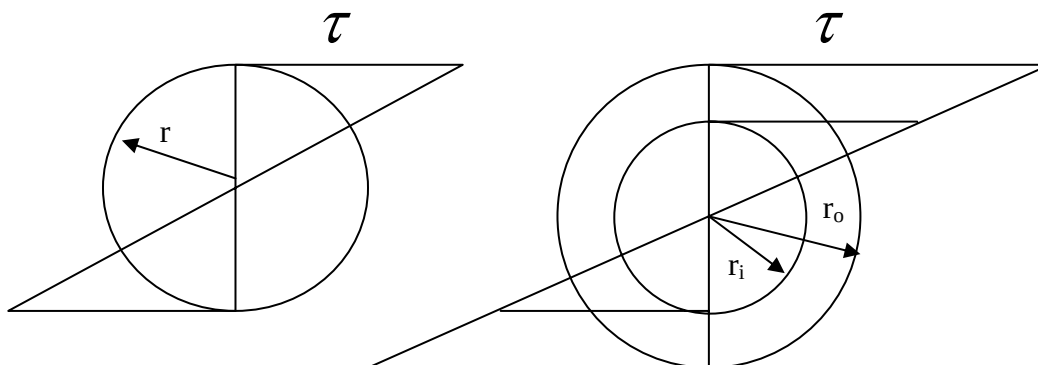
τ = Shear Stress [N/m²]

θ = Angle of Twist [radians]

L = Length of the Shaft [m]

γ = Shear Strain [radians]

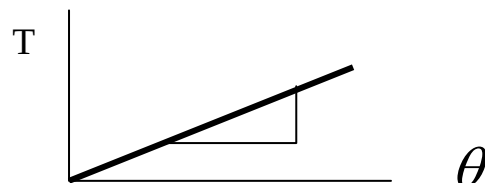
Comparison between solid and hollow shafts



Stiffness

- The Stiffness or Torsional Rigidity of the shaft is the slope of the graph of Torque against Angle of Twist. This is equivalent to the amount of torque required to produce a unit angle of twist.

$$K = \frac{T}{\theta} = \frac{GJ}{L}$$



- The strength of the shaft is a measure of the torque it can carry before reaching the maximum permissible shearing stress.

Power and Torque Relationship

In linear motion Power is give by

$$\text{Power} = \text{Force} \times \text{Linear Velocity} \quad \rightarrow \quad P = F \times V$$

Using similar analogy as in linear motion, shown above, the power transmitted due to angular motion of a shaft is give by its equivalent equation shown below.

$$\text{Power} = \text{Torque} \times \text{Angular Velocity} \quad \rightarrow \quad P = T \times \omega$$

Where $\omega = 2\pi\left(\frac{N}{60}\right)$ when N is the number of revolutions per minute (RPM).

Examples:

- (1) Determine the maximum stress in a hollow shaft when it is subjected to a twist of 5° over its entire length. What is the maximum torque the shaft can handle. Assume the shaft has a wall thickness of 5mm, is 2.5m long and has an internal diameter of 30mm.

Take G as 80GPa.

Hint:- Use $\frac{\tau}{r} = \frac{G\theta}{L} = \frac{T}{J}$

$$\tau = \frac{G \times \theta \times r}{L} = \frac{80 \times 10^9 \times \frac{5 \times \pi}{180} \times 35 \times 10^{-3}}{2.5}$$

$$\therefore \tau = 98 \text{ MPa}$$

$$T = \frac{J \times G \times \theta}{L} = \frac{\frac{\pi}{32} (0.04^4 - 0.03^4) \times 80 \times 10^9 \times \frac{5 \times \pi}{180}}{2.5}$$

$$\therefore T = 480 \text{ Nm}$$

- (2) A solid shaft is to transmit 720kW at 200 RPM. If the shaft is not to twist more than 1° over its entire length of 12m and the shear stress is not to exceed 45MPa, calculate the minimum shaft diameter required.

Take G as 80GPa.

Hint:-

use $\frac{G\theta}{L} = \frac{T}{J}$ & $P = T \times \omega$ to find r or d (1)

then use $\frac{\tau}{r} = \frac{T}{J}$ & $P = T \times \omega$ to find r or d (2)

Use judgment to decide what d should be.

From (1) $\frac{G\theta J}{L} = T$ and $T = \frac{P}{\omega}$

From (2) $\frac{\tau \times J}{r} = T$ and $T = \frac{P}{\omega}$

From (1) $J = \frac{P \times L}{G \times \theta \times \omega}$ and since $J = \frac{\pi}{32}(d^4)$

Therefore $d^4 = \frac{32 \times P \times L}{\pi \times G \times \theta \times \omega}$ or $d = \sqrt[4]{\frac{32 \times P \times L}{\pi \times G \times \theta \times \omega}}$

Where, $\theta = \frac{1 \times \pi}{180}$ & $\omega = 2 \times \pi \times \frac{200}{60}$

$$d = \sqrt[4]{\frac{32 \times 720 \times 10^3 \times 12}{\pi \times 80 \times 10^9 \times \frac{1 \times \pi}{180} \times 2 \times \pi \times \frac{200}{60}}}$$

$$d = 0.23m$$

The above value for d is one of the two options we have. The second option can be calculated by using (2)

From (2)
$$J = \frac{P \times r}{\tau \times \omega} = \frac{720 \times 10^3 \times \frac{d}{2}}{45 \times 10^9 \times 2 \times \pi \times \frac{200}{60}}$$

Since
$$J = \frac{\pi}{32} (d^4)$$

$$d^3 = \frac{32 \times 720 \times 10^3}{2 \times \pi \times 45 \times 10^9 \times 2 \times \pi \times \frac{200}{60}}$$

$$d = \sqrt[3]{\frac{32 \times 720 \times 10^3}{2 \times \pi \times 45 \times 10^9 \times 2 \times \pi \times \frac{200}{60}}}$$

$$d = 0.016m$$

This is the alternative value for d. The task now is to choose the right one.????

Tutorials

1. A solid aluminium shaft is to be replaced by a hollow steel shaft having the same outside diameter. If the shafts are to have the same torque-carrying capacity determine the inside diameter of the steel shaft as a percentage of the outside diameter if the permitted shear stress for the steel is twice that of the aluminium.

If the density of the steel is twice that of the aluminium express the weight of unit length of steel shaft as a percentage of that of the aluminium shaft.

(Answers. 84.1% ; 58.5%)

2. A solid steel shaft of 80 mm diameter is coupled to a hollow steel shaft of 100 mm outside diameter and 80 mm inside diameter. The length of the hollow shaft is 800 mm.

Determine the length of the solid shaft if the total angle of twist is 2° when the shafts are subjected to a torque of 10 kNm. Take the Modulus of Rigidity for steel, $G = 80 \text{ GN/m}^2$.

(Answer: 568 mm)

3. A shaft is to transmit a mean power of 2 MW when running at 120 rev/min. The maximum allowable shear stress in the shaft material is 60 MN/m^2 and the torque on the shaft will fluctuate, the maximum torque being 1.2 times the mean torque. Determine:-

- (a) the required diameter if a solid shaft is used;
- (b) the dimensions of a suitable hollow shaft if its internal diameter is to be 0.8 times the external diameter;
- (c) the corresponding angle of twist of the solid shaft if its length is 5 m. Take $G = 80 \text{ GN/m}^2$ throughout.

(Answers: (a). 253.1mm (b) o/d = 301.7mm i/d = 241.3mm (c) 1.698°)

4. A tubular steel propeller shaft has an internal diameter, which is three-quarters of the external diameter. The angle of twist of the shaft is 0.5° per metre length of shaft when the maximum shear stress is 80 MN/M^2 .

Determine the external diameter of the shaft and the maximum power which may be transmitted by the shaft at a speed of 250 rev/min. Take $G = 80 \text{ GN/m}^2$.

(Answers 229.2mm 3.385 MW)

5. If it is specified that the angle of twist of a shaft is not to exceed 1° over a length of shaft equal to 20 times its diameter, show that the maximum shear stress in the shaft depends only on its modulus of rigidity. If such a shaft

is of steel having a modulus of rigidity of 80 GN/m^2 , is solid, and has a diameter of 80 mm, determine.

- (a) the maximum Torsional shear stress that could be transmitted by the shaft;
- (b) the corresponding maximum torque and power that can be transmitted at 300 rev/min.

(Answers. (a) 34.9 MN/m^2 (b) 3509 Nm; 110.2 kW)

6. The propeller shaft of a motor vehicle consists of a tube of external diameter 45 mm with solid end sections of diameter 40 mm. The length of the tubular section is 800 mm and that of each solid section is 50 mm. The maximum shear stress in the solid sections is to be 70 MN/m^2 . Determine:-

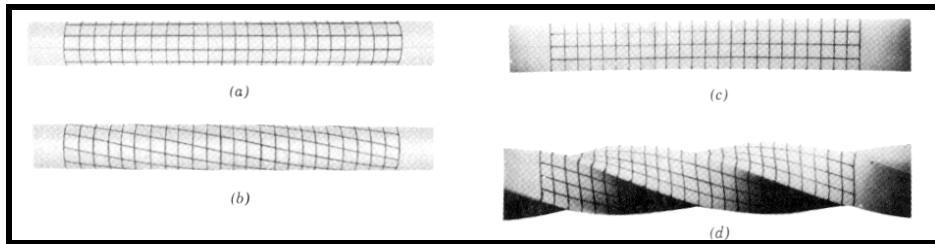
- (a) the internal diameter of the tubular section so that it will have the same maximum shear stress as the solid sections;
- (b) the total angle of twist of the propeller shaft produced under the maximum stress conditions. Take G , as 80 GN/m^2 throughout

(Answers. (a) 33.2 mm (b) $0.0355 \text{ rad} = 2.03^\circ$)

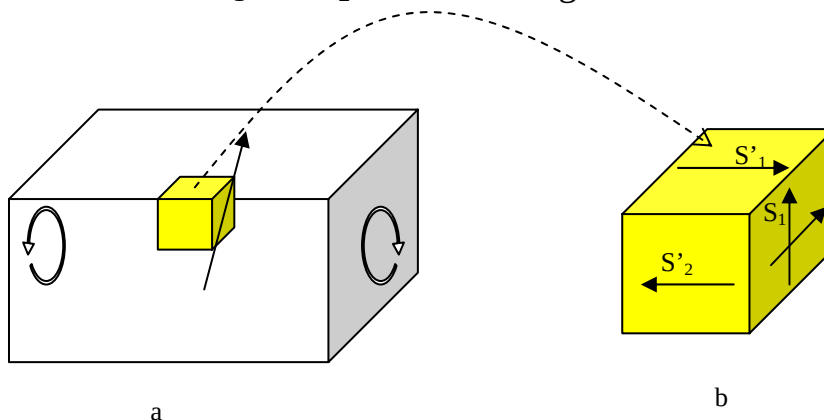
7. A hollow shaft has an external diameter of 160 mm and an internal diameter of 120 mm. If the maximum shear stress is not to exceed 50 MN/m^2 , determine the maximum power that could be transmitted by the shaft when rotating at 180 rev/min, and the angle of twist (in degrees) over a 4 m length when transmitting this power. Take G as 83 GN/m^2 throughout.

(Answers: 518 kW; 1.726°)

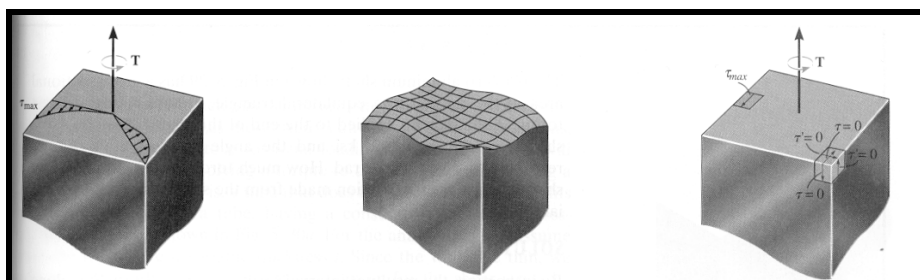
Torsion of non-circular section beams

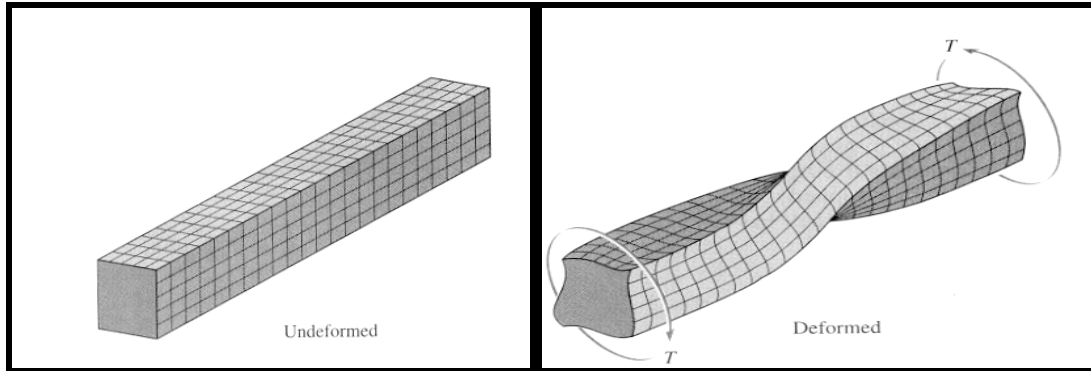


Until 1820, it was thought that the shearing stresses in any torsionally loaded member were proportional to the distance from its axis, as it is in the case of circular section beams. It was proved experimentally that this is not true for rectangular cross sections. An examination of the figure below will verify this conclusion. If the stresses in the rectangular bar were proportional to the distance from its axis, the maximum stress would occur at the corners. However, if there was a stress of any magnitude at the corner, it could be resolved into the components. If these components existed, the two components shown dashed would also exist. These last components cannot exist, since the surfaces on which they are shown are free boundaries. Therefore, the shearing stresses at the corners of the rectangular bar must be zero. Therefore S_1 and S_2 have zero magnitude.



In general, ***except for members with circular cross sections***, every section will warp when the bar is twisted.





For the case of the rectangular bar shown, the distortion of the small squares is maximum at the midpoint of a side of the cross section and disappears at the corners. Since this distortion is a measure of shearing strain, Hooke's law requires that the shearing stress is also maximum at the midpoint of a side of the cross section and zero at the corners.

To summarise:

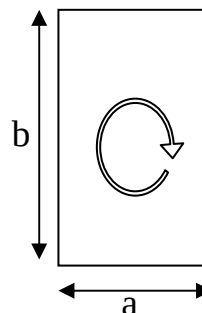
- Plane sections do not remain plane.
- Shear stresses are not proportional to distance from the centre of twist.
- Shear stress at corners of the rectangular section is zero.
- The distortion due to warping is maximum at mid sides and zero at corners.
- As a consequence, maximum shear stresses also accure at mid-sides and are zero at corners.

Equations for the maximum shearing stress and angle of twist for a rectangular section obtained using Saint Venant's theory are

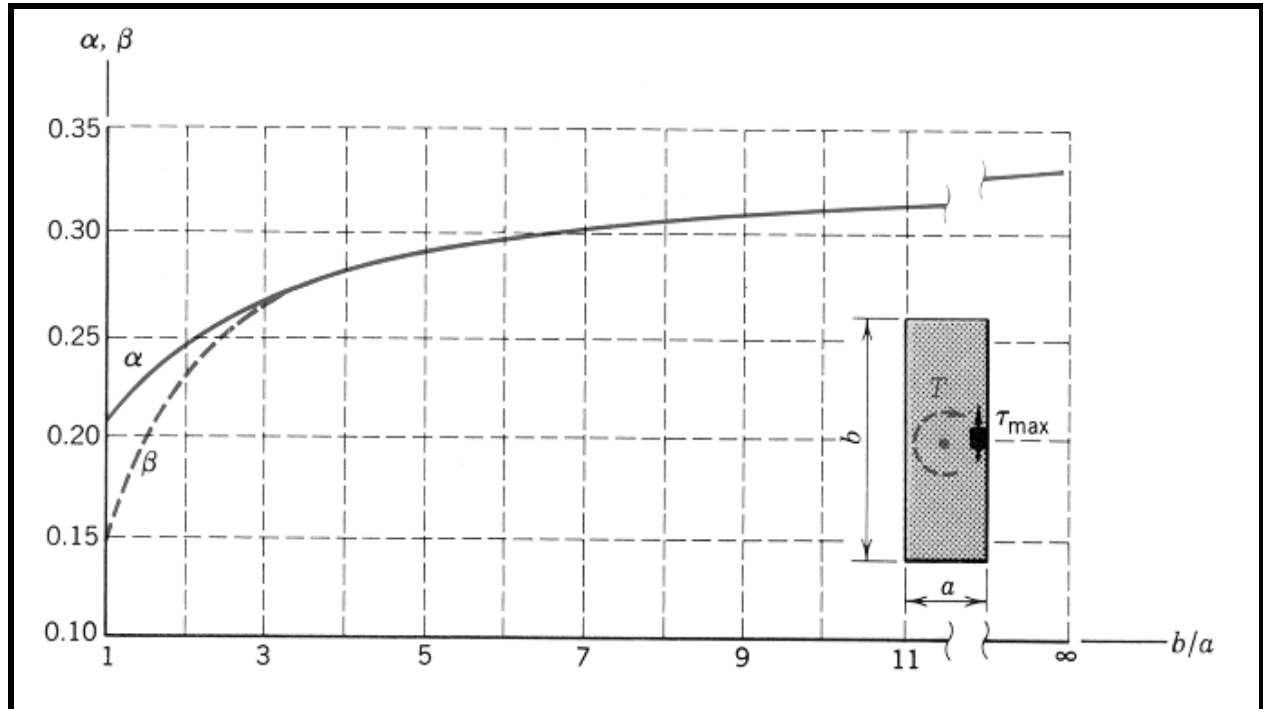
$$\tau_{Max} = \frac{T}{\alpha a^2 b}$$

and

$$\theta = \frac{TL}{\beta a^3 b G}$$



where a and b are the lengths of the short and long side of the rectangle respectively. The values of α and β can be obtained from the table or the graph shown below.

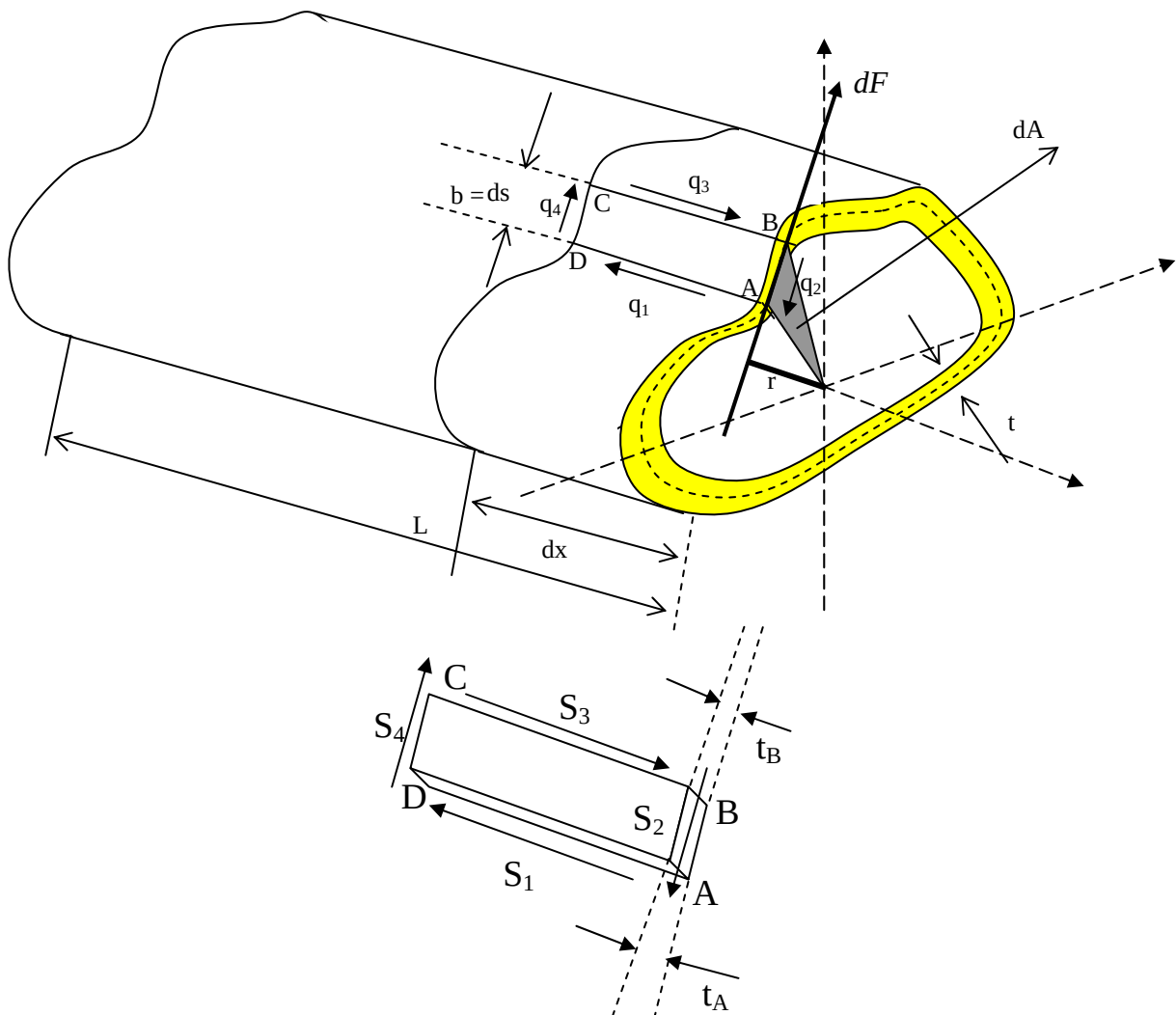


$\frac{b}{a}$	α	β
1.0	0.208	0.1406
1.2	0.219	0.1661
1.5	0.231	0.1958
2.0	0.246	0.229
2.5	0.258	0.249
3.0	0.267	0.263
4.0	0.282	0.281
5.0	0.291	0.291
10.0	0.312	0.312
∞	0.333	0.333

Torsion of thin-walled non-circular tube

Another class of section to be analysed is the thin walled non-circular section. Which can be readily analysed by elementary methods. A typical section is illustrated below which represents a noncircular section with a wall of variable thickness t .

A useful concept associated with the analysis of thin-walled sections is shear flow q , defined as internal shearing force per unit of length of the thin section. Typical unit for q is Newton per meter. In terms of stress, q equals $\tau(l)(t)$, where τ is the average shearing stress across the thickness t . It will be demonstrated that the shear flow on a cross-section is constant even though the thickness of the section wall varies.



To do that we consider the equilibrium of the block element ABCD of the member when it is subjected to pure torsion. The above diagram shows the block ABCD cut

from the member between points A and B, and the member is subjected to pure torsion.

The shear forces $s_1, \dots, s_4$ alone (no normal forces) are necessary for equilibrium. Summing forces in the x direction gives:

$$s_1 = s_3$$

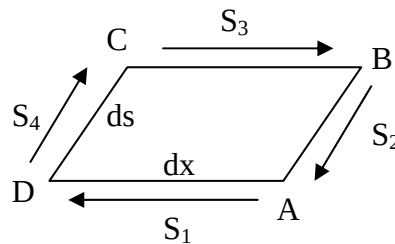
$$q_1 dx = q_3 dx$$

From which

$$q_1 = q_3$$

Similarly

$$q_2 = q_4$$

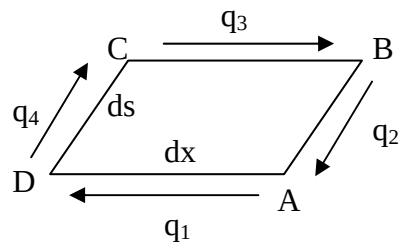


- Considering the equilibrium of element ABCD
- q_1 on the face AD is the same as q_3 on the face BC
- q_2 on the face AB is the same as q_4 on the face CD.
- Taking moment about any point, say point A.

$$q_3 \times dx \times ds = q_4 \times ds \times dx$$

$$\therefore q_3 = q_4$$

$$\therefore q_1 = q_2 = q_3 = q_4$$



i.e. shear flow is constant at any point on the cross section of the tube.

Relationship between T and q

It is now possible to develop an expression for the resulting torque on the section. To do that we must consider the force dF acting through the centre of a differential length of perimeter ds , as shown above. The resisting torque T , is the resultant of the moments of the internal forces dF around the perimeter of the section.

External Torque T = Internal moment of resistance

$$T = \oint q.ds.r$$

Since $q = \tau.t$ constant.

$$\therefore T = \tau.t \oint r ds$$

but $r.ds$ is twice the area of the triangle OAB

$$\therefore \oint r ds = 2A$$

where A is area enclosed by the median line.

$$\therefore T = \tau.t.2.A$$

or

$$\tau = \frac{T}{2At}$$

Notice that τ is maximum when t is minimum.

Alternatively

$$q = \frac{T}{2A}$$

Relationship between T and θ

There are many methods to develop this but the simplest method is to use strain energy. Strain energy due to torsion must be equal to work done by torque T causing an angular displacement θ . The Strain energy per unit volume, U, due to shear stress τ is given by

$$\therefore U = \frac{\tau^2}{2G} \times \text{Volume}$$

$$\therefore U = \oint \frac{\tau^2}{2G} \times L \times t \times ds \quad \text{Since} \quad \tau \times t = q$$

$$\therefore U = \oint \frac{q^2}{2G} \times L \times \frac{ds}{t}$$

but

$$\frac{q^2 L}{2G} = \text{Constant}$$

$$\therefore U = \frac{q^2 L}{2G} \oint \frac{ds}{t}$$

But from above $q = \frac{T}{2A}$

$$\therefore U = \frac{T^2 L}{8A^2 G} \oint \frac{ds}{t}$$

But $U = \frac{1}{2} T \theta$ which is the work done by T

$$\frac{1}{2}T\theta = \frac{T^2 L}{8A^2 G} \oint \frac{ds}{t}$$

$$\theta = \frac{TL}{4A^2 G} \oint \frac{ds}{t}$$

In practice the above equations can be applied in the following form.

$$\theta = \frac{TL}{4A^2 G} \sum_{i=1}^n \frac{s_i}{t_i}$$

Alternatively, since $q = \frac{T}{2A}$

$$\frac{\theta}{L} = \frac{q}{2AG} \oint \frac{ds}{t}$$

In practice the above equations can be applied in the following form.

$$\frac{\theta}{L} = \frac{q}{2AG} \sum_{i=1}^n \frac{s_i}{t_i}$$

or

$$\frac{\theta}{L} = \frac{1}{2AG} \sum_{i=1}^n \frac{q_i s_i}{t_i}$$

If thickness t is constant throughout, $\sum s$ will be the length of the perimeter of the cross section bound by median line.

$$\theta = \frac{TLs}{4A^2 Gt} \quad \text{or} \quad \theta = \frac{qLs}{2AGt}$$

$J = \frac{4A^2t}{s}$ = is referred to as torsional constant which is not the same as the polar second moment of area.

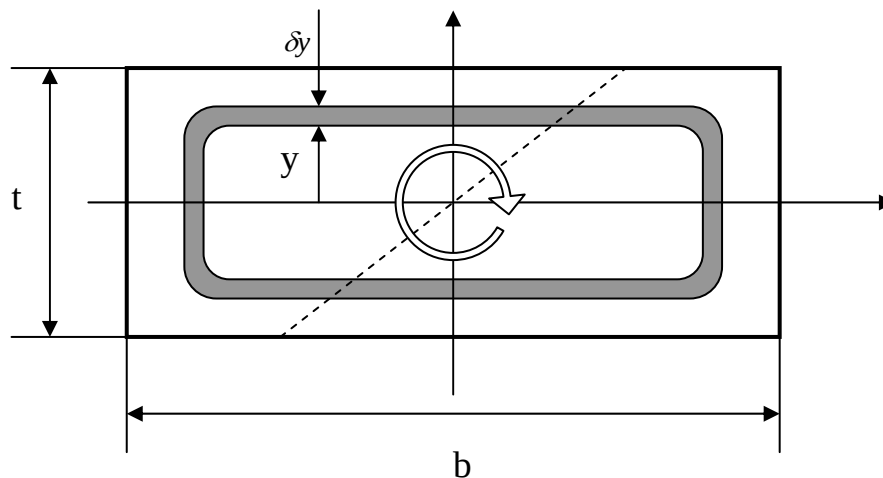
Therefore:

$$\boxed{\theta = \frac{TL}{GJ}} \quad \text{or} \quad \frac{T}{J} = \frac{G\theta}{L} \quad \text{which appears to look similar to that of the}$$

equation for torsion of circular section except that J is the torsional constant.

Torsion of Thin Rectangular Strip

An approximate solution for the torsion of a strip of rectangular cross section whose thickness is small compared with the width, shown below, can be obtained by considering the strip to be built up of a series of thin walled concentric tubes which twist by the same amount.



Assumption here is that $\frac{b}{t} \geq 5$

Consider the above rectangular strip. Here the analysis is similar to the analysis of thin walled rectangular tube, shown shaded, integrated through the cross section.

The equation $\tau = \frac{T}{2At}$ when applied to the above shaded ring/strip results in;

$$\tau = \frac{\delta T}{2 \times b \times 2y \times \delta y}$$

or

$$\frac{\delta T}{\delta y} = 4 \times b \times y \times \tau$$

but $\theta = \frac{TLs}{4A^2Gt}$ and when applied to the above shaded strip

$$\theta = \frac{\delta T \times L \times (2b + 4y)}{4 \times 4b^2y^2 \times G \delta y}$$

But since $2b \geq 4y$

Then

$$\theta = \frac{\delta T}{\delta y} \times \frac{L}{8by^2G}$$

or

$$\frac{\delta T}{\delta y} = \frac{8by^2G\theta}{L}$$

By equating the above with equation $\frac{\delta T}{\delta y} = 4 \times b \times y \times \tau$ we get

$$4 \times b \times y \times \tau = \frac{8by^2G\theta}{L}$$

Or

$$\tau = \frac{2Gy\theta}{L}$$

ie. τ varies linearly with y and is maximum at $y = \frac{t}{2}$

$$\therefore \tau_{\max} = Gt_{\max} \left(\frac{\theta}{L} \right)$$

Relationship between T and θ

Previously we derived that

$$\delta T = 4 \times b \times y \times \tau \times \delta y$$

Since $\tau = \frac{2Gy\theta}{L}$

$$\delta T = 4 \times b \times y \times 2G \times y \times \frac{\theta}{L} \delta y$$

$$\delta T = 8 \times b \times G \frac{\theta}{L} \times y^2 \delta y$$

Integrating the above through the thickness:

$$T = 8 \times b \times G \frac{\theta}{L} \int_0^{\frac{t}{2}} y^2 dy$$

integration results in

$$T = G \left(\frac{bt^3}{3} \right) \left(\frac{\theta}{L} \right)$$

where

$$\frac{bt^3}{3} = J \quad \text{the torsional constant}$$

Torsion of thin walled open section beams

$$\tau_{\max} = Gt_{\max} \left(\frac{\theta}{L} \right)$$

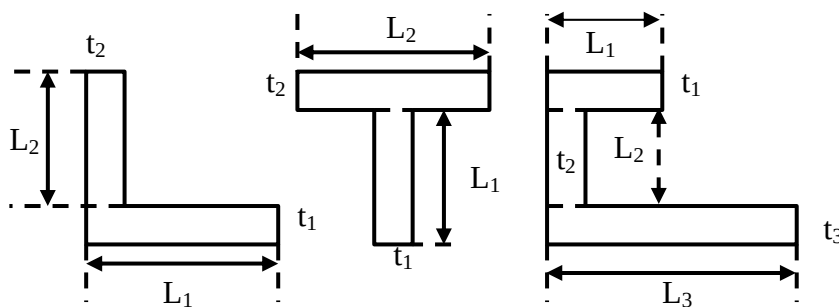
This theory can be used for built up sections such as I, L, T, C etc. sections where

$\frac{\theta}{L}$ is the same for all segments

$$T = GJ \left(\frac{\theta}{L} \right)$$

where J is calculated as shown in following examples.

$$J = \frac{L_1 t_1^3}{3} + \frac{L_2 t_2^3}{3} + \frac{L_3 t_3^3}{3}$$



Example 1

A rectangular box section aluminium alloy has outside dimensions 100*50mm. The plate thickness is 2mm for the 50mm sides and 3mm for the 100mm sides. If the $\tau_{All} \leq 95\text{Mpa}$, determine the T max that can be applied.

Solution

Max stress occur at the thinnest plate

$$q = \tau \times t = 95 \times 10^6 \times 2 \times 10^{-3} = 190 \times 10^3 \text{ N/m}$$

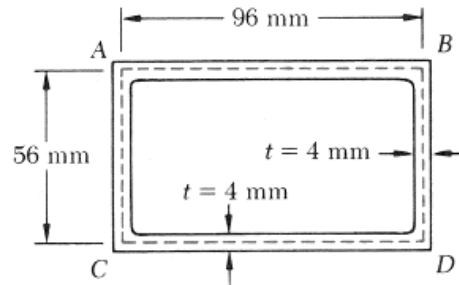
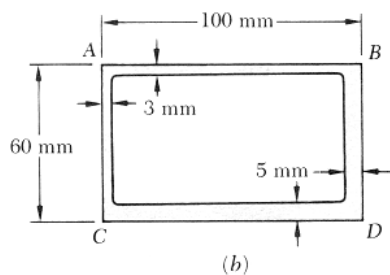
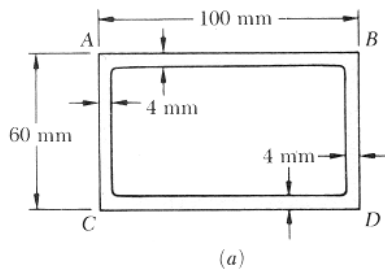
The torque that can be transmitted by this section is

$$T = 2 q A = 2 \times 190 \times 10^3 \times (100-2)(50-3) \times 10^{-6} = 1750\text{Nm}$$

Example 2

Rectangular cross section of 60*100mm Aluminium tube is subjected to a torque of 3kNm. Determine the shearing stresses at each of the four walls when assuming;

- Uniform 4mm wall thickness.
- Wall AB and AC are 3mm thick and BD and CD are 5mm thick.



Solutions

a) For uniform wall thickness

$$A = (100-4)(60-4) = 5.376 \times 10^{-3} \text{ m}^2 \text{ area bounded by the median.}$$

From the notes we have:

$$\tau = \frac{T}{2At} = \frac{3 \times 10^3 \text{ Nm}}{2 \times (4 \times 10^{-3} \text{ m})(5.376 \times 10^{-3} \text{ m}^2)} = 69.8 \text{ MPa.}$$

b) The area bounded by the median is the same as above. Using $t = 3$ and $t = 5$ in the above equation gives:

$$\tau_{AB} = \tau_{AC} = \frac{T}{2At} = \frac{3 \times 10^3 \text{ Nm}}{2 \times (3 \times 10^{-3} \text{ m})(5.376 \times 10^{-3} \text{ m}^2)} = 93.0 \text{ MPa.}$$

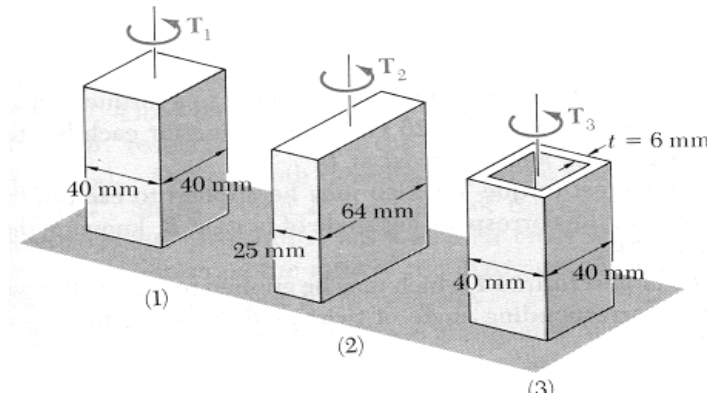
$$\tau_{BD} = \tau_{CD} = \frac{T}{2At} = \frac{3 \times 10^3 \text{ Nm}}{2 \times (5 \times 10^{-3} \text{ m})(5.376 \times 10^{-3} \text{ m}^2)} = 55.8 \text{ MPa.}$$

Shear stress only depends on thickness

Example 3

Using Maximum allowable shear stress $\tau_{All} = 40 \text{ MPa}$, determine the largest torque which may be applied to each of the bars and to the brass tubes shown below.

Note: - Two solid bars have the same cross section area, the square bar and tube have the same outside dimension.



Solution

a) Bar with square cross section

$$a = b = 0.04 \text{ m}, \quad b/a = 1 \quad \therefore \alpha = 0.208$$

$$\tau_{\max} = \frac{T}{\alpha a^2 b} = 40 \text{ MPa} = \frac{T_1}{0.208 (0.04)^3}$$

$$\therefore T_1 = 532 \text{ Nm}$$

b) Bar with rectangular cross section

$$b = 0.064, \quad a = 0.025, \quad b/a = 2.56 \quad \therefore \alpha = 0.259$$

$$\tau_{\max} = \frac{T}{\alpha a^2 b} = 40 \text{ MPa} = \frac{T_2}{0.259 (0.064)(0.025)^2}$$

$$\therefore T_2 = 414 \text{ Nm}$$

c) Square tube

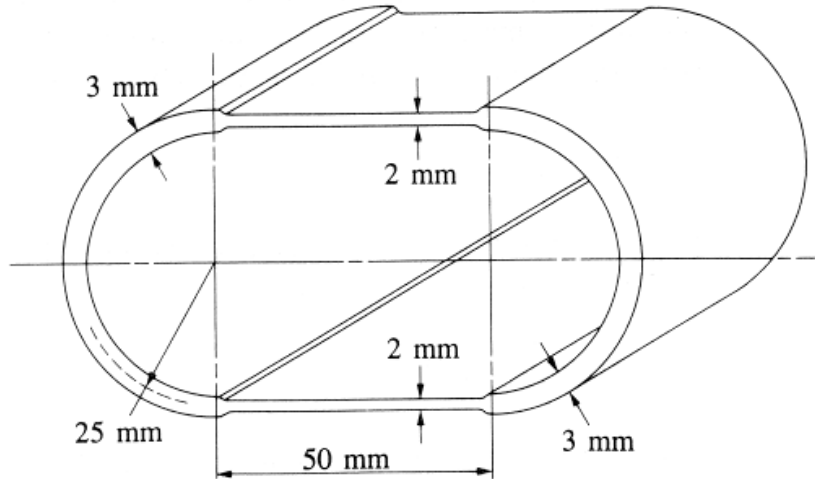
$$A = (0.034) \times (0.034) = 1.156 \times 10^{-3} \text{ m}^2$$

$$\tau = \frac{T}{2At} = 40 = \frac{T_3}{2(0.006)(1.156 \times 10^{-3})}$$

$$\therefore T_3 = 555 \text{ Nm}$$

Example 4

The light alloy stabilising strut of a high wing monoplane is 2 m long and has cross section shown below. Determine the torque that can be sustained and the angle of twist if the $\tau_{\max}$ is 28Mpa and $G = 27\text{Gpa}$.



Solution

The area enclosed by the median line of the wall thickness is

$$A = (\pi \times 25^2) + (50 \times 50) = 4460 \text{ mm}^2$$

To find maximum torque we use minimum thickness as criteria because shear stress is maximum there.

$$T = 2At\tau$$

$$T = 2 * 4460 \times 10^{-6} \times 2 \times 10^{-3} \times 28 \times 10^6 = 500 \text{ Nm}$$

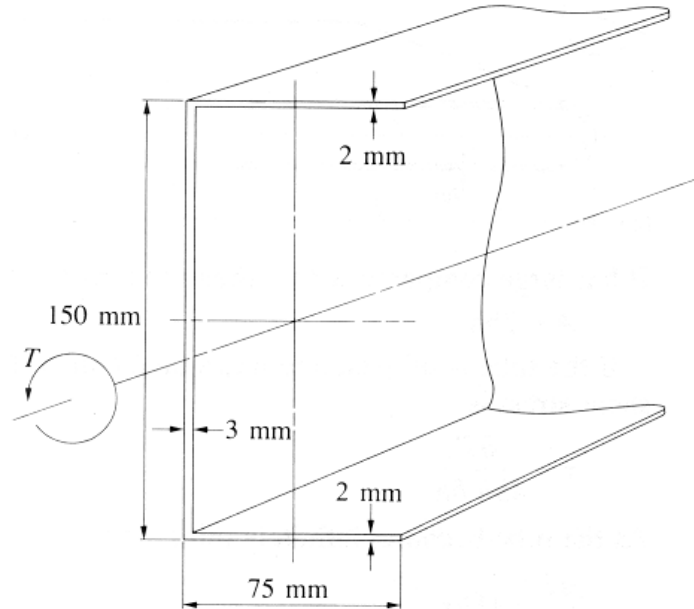
Total angle of twist can be calculated using:

$$\theta = \frac{TL}{4A^2G} \oint \frac{ds}{t}$$

$$\theta = \frac{500 * 2}{4 \times (4460)^2 \times 10^{-12} \times 27 \times 10^9} \times \left[\frac{100}{2} + \frac{50\pi}{3} \right] = 0.048 \text{ rad}$$

Example 5

Determine the angle of twist per unit length and the maximum shear stress in the aluminum channel shown below when subjected to a pure torque of 20 Nm. $G = 27 \text{ GPa}$.



Solution

This channel is analysed as 3 rectangular strips, using equation developed previously.

$$T = G \left(\frac{bt^3}{3} \right) \left(\frac{\theta}{L} \right)$$

$$T_1 = \frac{1}{3} \times 75 \times 10^{-3} \times (2 \times 10^{-3})^3 \times 27 \times 10^9 \times \frac{\theta}{l} = 5.4 \frac{\theta}{l}$$

$$T_2 = \frac{1}{3} \times 150 \times 10^{-3} \times (3 \times 10^{-3})^3 \times 27 \times 10^9 \times \frac{\theta}{l} = 36.5 \frac{\theta}{l}$$

$$\text{but } T_{\text{tot}} = 2T_1 + T_2 = 20 \quad \therefore 47.3 \frac{\theta}{l} = 20$$

$$\frac{\theta}{l} = 0.422 \text{ rad / m} = 24.2^\circ / \text{m}$$

The maximum shear stress in the flange is given by $\tau_{\text{max}} = G t_{\text{max}} \left(\frac{\theta}{L} \right)$

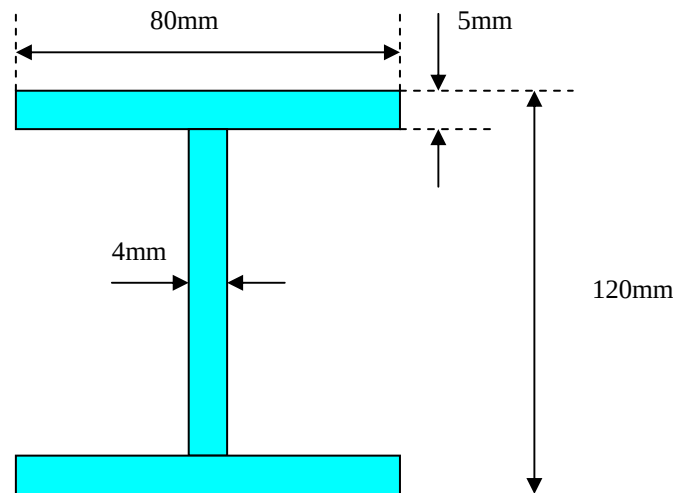
$$\tau_{\text{max}} = 2 \times 10^{-3} \times 27 \times 10^9 \times 0.422 = 22.8 \text{ MN / m}^2$$

in the web

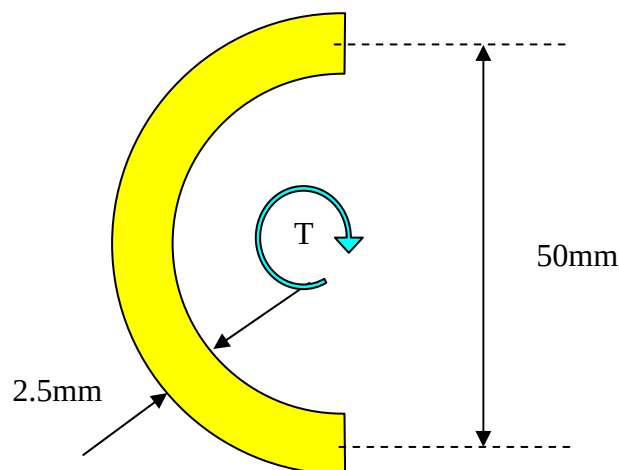
$$\tau_{\text{max}} = 3 \times 10^{-3} \times 27 \times 10^9 \times 0.422 = 34.2 \text{ MN / m}^2$$

Tutorials

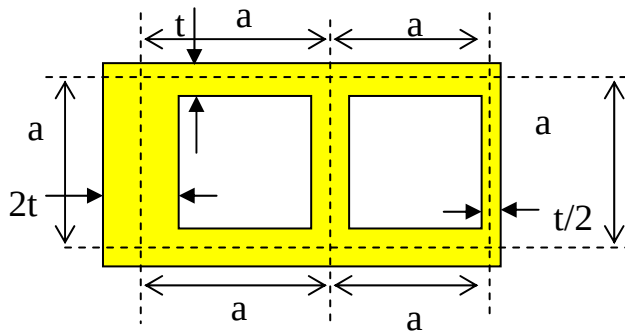
- 1) Steel I-section, shown below is subjected to a torque T . Find the maximum value of T if the shear stress is limited to 50N/mm^2 and the twist per meter length must not exceed 6° . The shear modulus $G = 80\text{ kN/mm}^2$



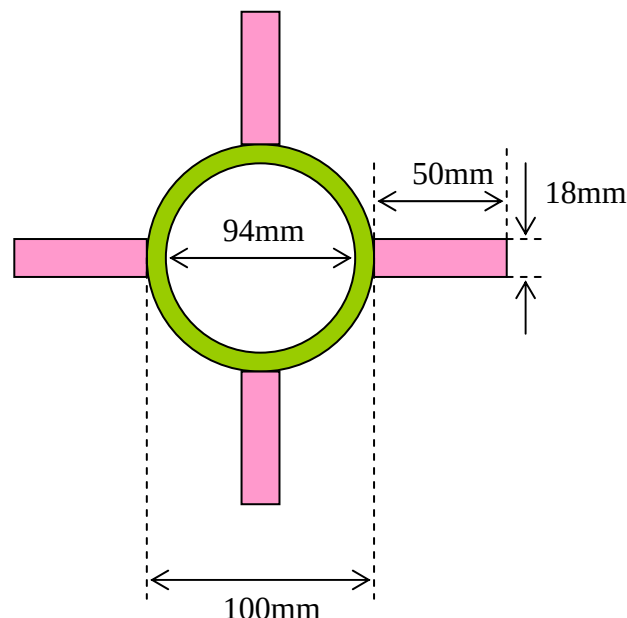
- 2) An extruded section in light alloy is in the form of a semicircular tube, 50mm mean diameter and 2.5mm thick. If a torque is applied to the section and the angle of twist is to be limited to 4° per meter length of tube, estimate the torque and maximum shear stress. Take $G = 27\text{kN/mm}^2$



- 3) A 2 cell baggage compartment of an aircraft is shown below with dimensions $a = 150$, $t = 12\text{mm}$, $G = 38\text{GPa}$. Find the maximum torque that can be transmitted if $\tau_{\max} = 35\text{MPa}$. Assume sides have lengths 'a' and thickness $2t$, t and $t/2$ at various locations shown on the diagram. Assume thickness t on side where thickness is not indicated.



- 4) A Rocket nozzle also acts as a torsion member and has a cross section shown below. The tube has inner and outer diameters of 94 and 100 mm respectively. Each fin is treated as strip of 50mm by 18mm, and the whole thing is 3 m long. If max stress is not to exceed 56MPa, calculate the maximum torque, which can be carried by the member and the resulting angle of twist over the entire length of this member. Assume $G = 82\text{GPa}$.



- 5) A thin-wall torsion member which, forms an integral part of the wing of an aircraft, is subjected to torque T . The member has the cross section shown in Figure a. The wall thickness t is constant throughout the section. Given that for a thin noncircular closed section beam:

$$T_i = 2A_i q_i, \quad \text{where } q_i = \tau * t$$

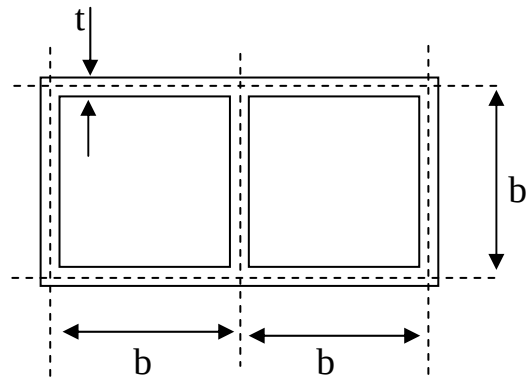
And

$$\theta = \frac{\tau L S}{2AG}$$

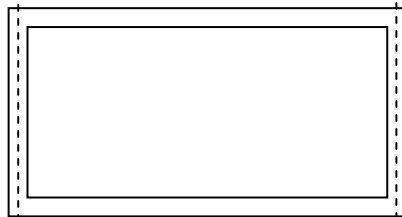
- a) Show that the shear stress in the outer walls of the section shown in Figure below. is given by $\tau = \frac{T}{4b^2 t}$
- b) Show that the shear stress in the internal vertical web is zero.
- c) Show that the unit angle of twist is given by $\theta = \frac{3T}{8b^3 Gt}$.
- d) It was recommended that the internal web should be removed, as it does not contribute to the torsional stiffness of the section. With interior web removed, as shown in Figure b, show that the shear stress in the outer wall and the unit angle of twist are the same as those shown above.
- e) If as a result of the web removal a length wise crack is appeared through a significant length of the member, forming an open cross section as in Figure c, discuss, with the help of some derivation, the effect of this defect on the performance of the bar in the terms of the unit angle of twist and the shear stress distribution.

In all cases assume constant thickness and disregard the effects of stress concentrations at interior corners.

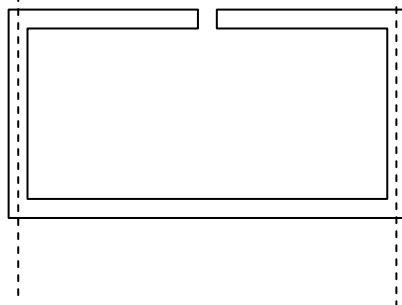
(a)



(b)



(c)



Assume constant thickness t throughout

- 6) As part of a design for a new aircraft engine propeller system, there is a requirement to mould the propeller to a composite torsion member shown in figure a. The tube has inner and outer diameters of 90 and 100 mm respectively and the blades, which run along the entire 200mm length of the tube, have a width of 100mm and thickness of 25mm. If the maximum stress is not to exceed 56MN/m^2 , calculate;

- i) The maximum torque that can be carried by the torsion member.
- ii) The unit angle of twist.

If a length-wise crack appears on the tube and runs through the significant length of the tube as in figure b;

- iii) Discuss with the help of some calculations the effect of this crack on the maximum torque and the unit angle of twist for this propeller.

Assume $G=82\text{GPa}$.

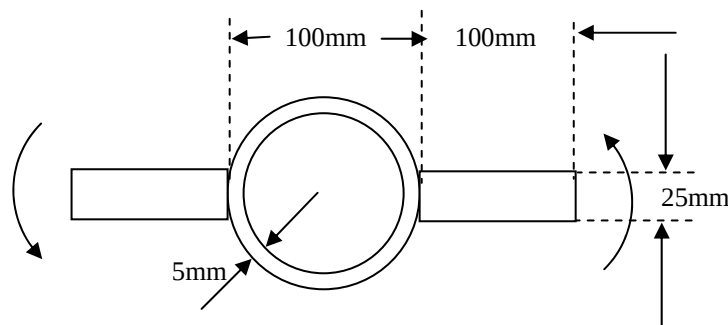


Figure a

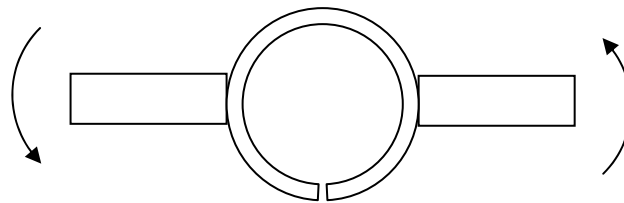
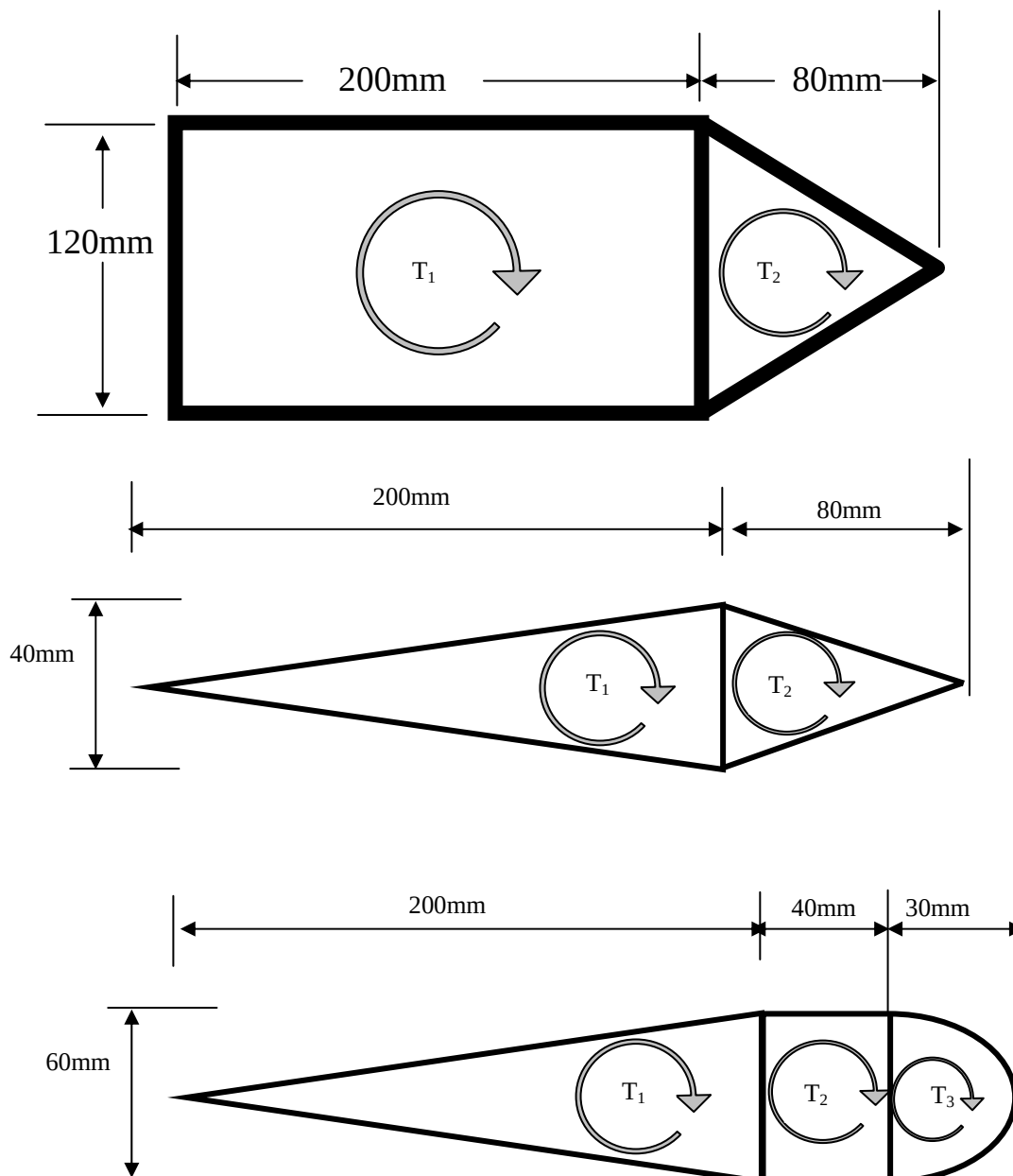


Figure b

- 7) A thin-wall torsion member, which forms an integral part of a wing structure of an aircraft, is made out of extruded aluminium. It is 3 m long and is subjected to total torque $T = T_1 + T_2 = 11 \text{ kNm}$. The member has the cross section as shown below. The beam has a uniform wall thickness $t = 5 \text{ mm}$ except for the internal web which is 4 mm thick. Determine;

- The maximum shear stress in the section.
- The total angle of twist.

In all cases disregard the effect of stress concentrations at interior corners.



- 8) A thin-wall torsion member which, forms an integral part of the structure of a wing assembly for a commercial airliner, is formed by super plastic forming and is subjected to torque $T = 1.2 \text{ kNm}$. The member has the cross section shown in Figure (a). The wall thickness $t = 2 \text{ mm}$ and is constant throughout the section.
- Calculate the shear stress in the section
 - Calculate the angle of twist per unit length.
 - If a length-wise crack appears through a significant length of the member, forming an open cross section as shown in Figure (b), calculate the angle of twist per unit length and the shear stress in the system.

In all cases disregard the effects of stress concentrations at interior corners. Assume symmetry about YY axis.

